

BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

Complete

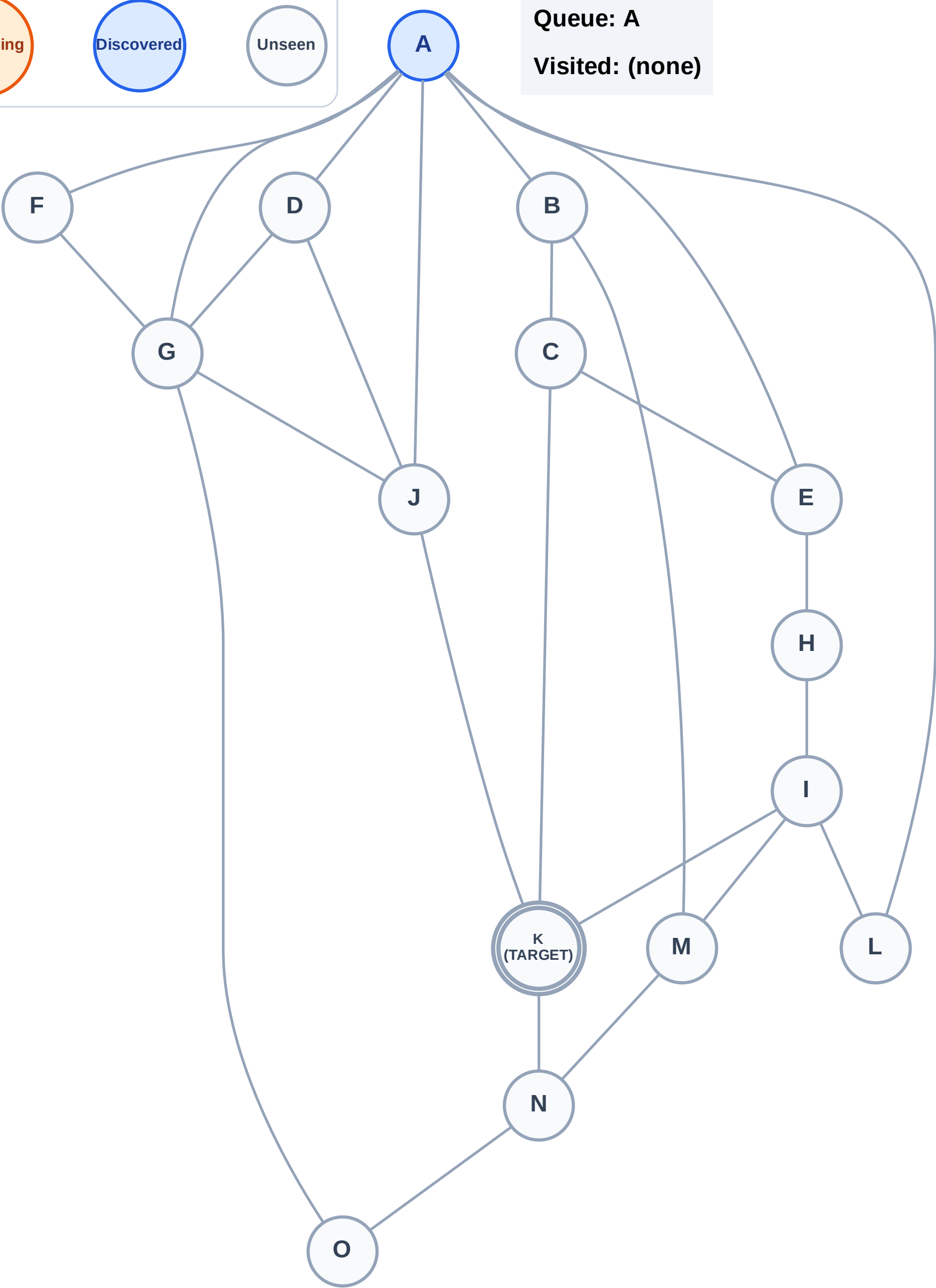
Processing

Discovered

Unseen

Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

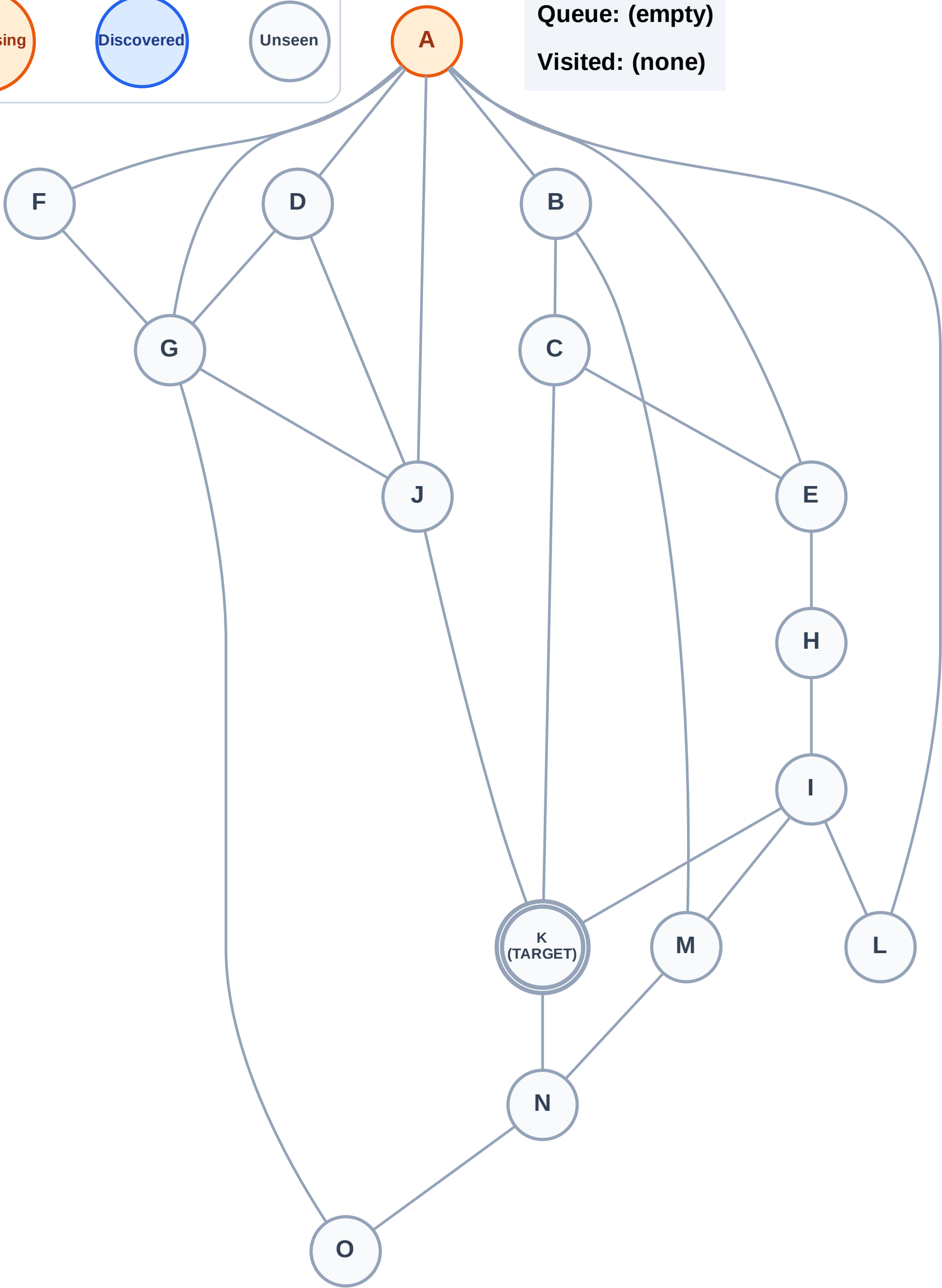
Complete

Processing

Discovered

Unseen

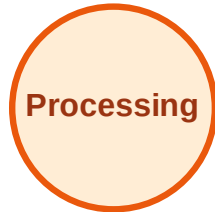
Queue: (empty)
Visited: (none)



BFS Traversal

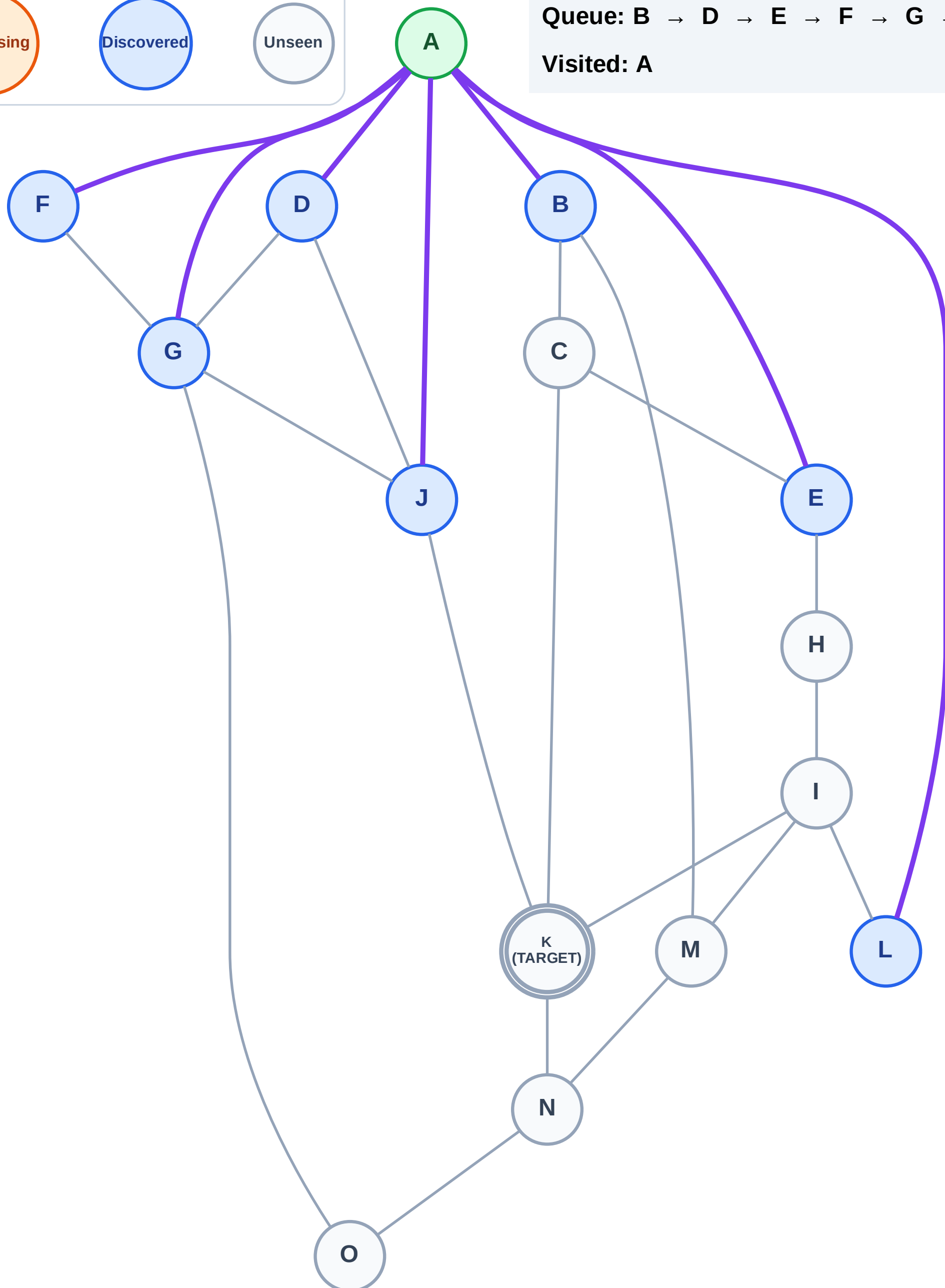
Step 3: Discover B, D, E, F, G, J, L from A

Legend



Queue: B → D → E → F → G → J → L

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

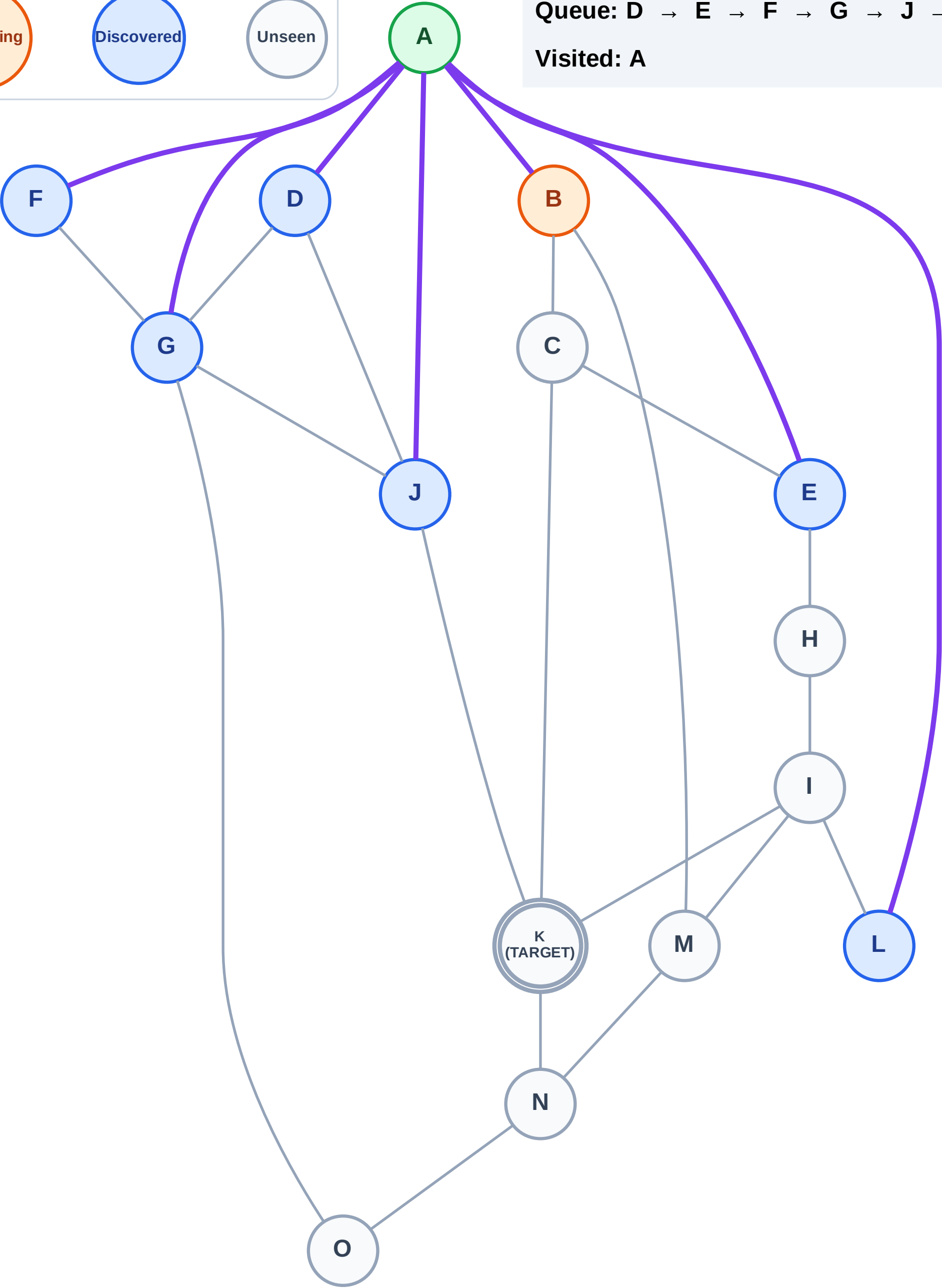
Processing

Discovered

Unseen

Queue: D → E → F → G → J → L

Visited: A



BFS Traversal

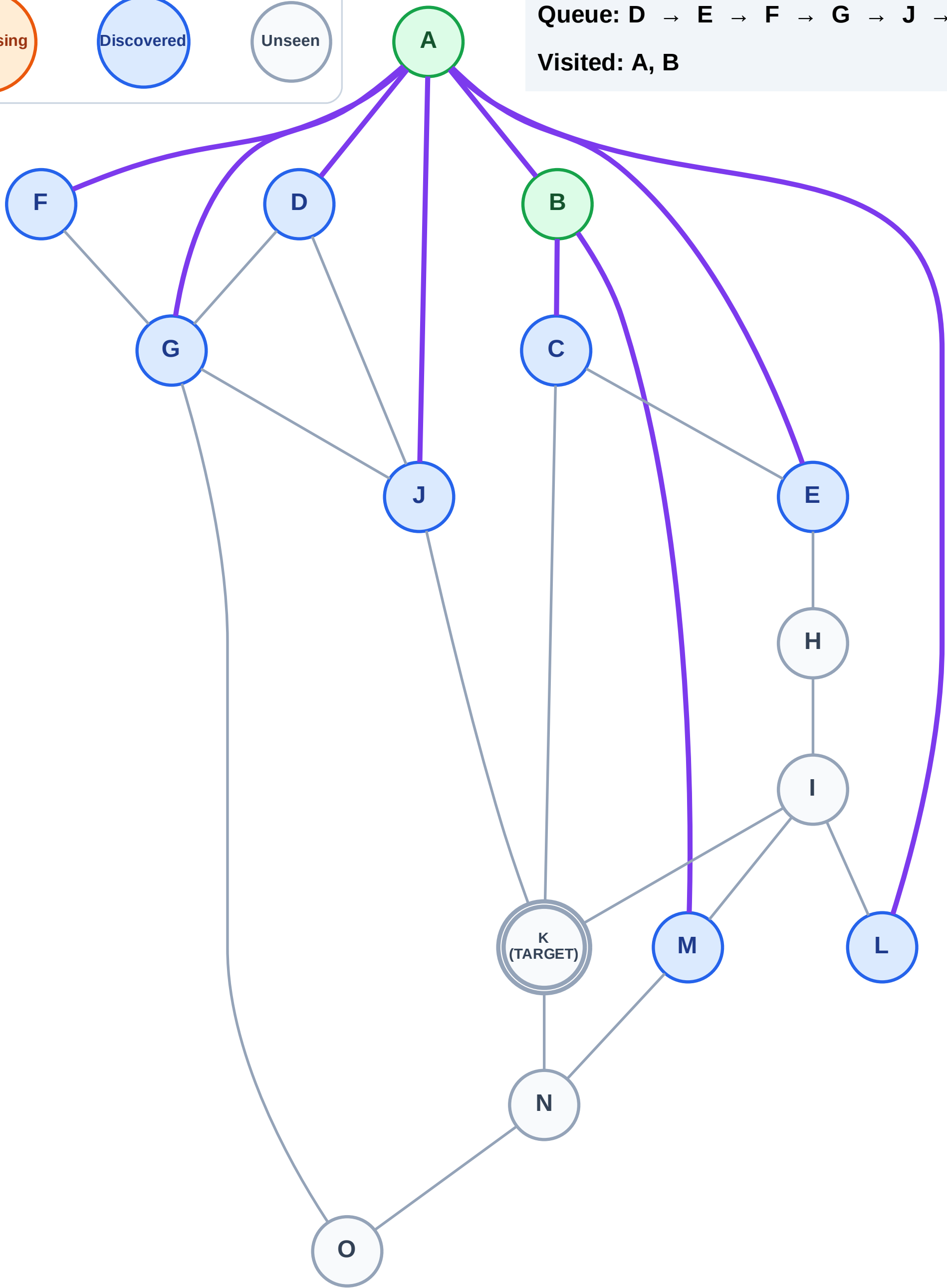
Step 5: Discover C, M from B

Legend



Queue: D → E → F → G → J → L → C → M

Visited: A, B



BFS Traversal

Step 6: Process node D

Legend

Complete

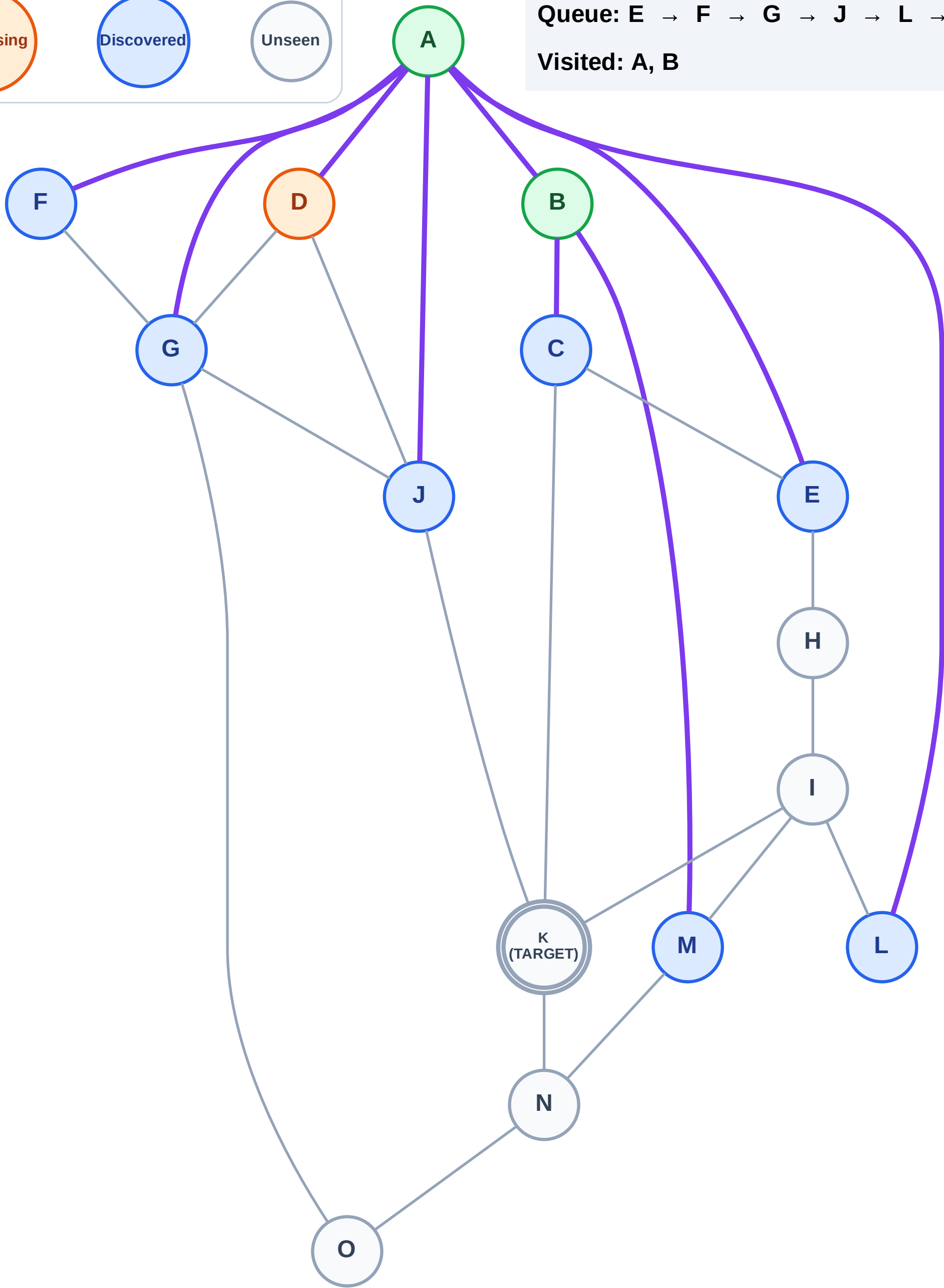
Processing

Discovered

Unseen

Queue: E → F → G → J → L → C → M

Visited: A, B



BFS Traversal

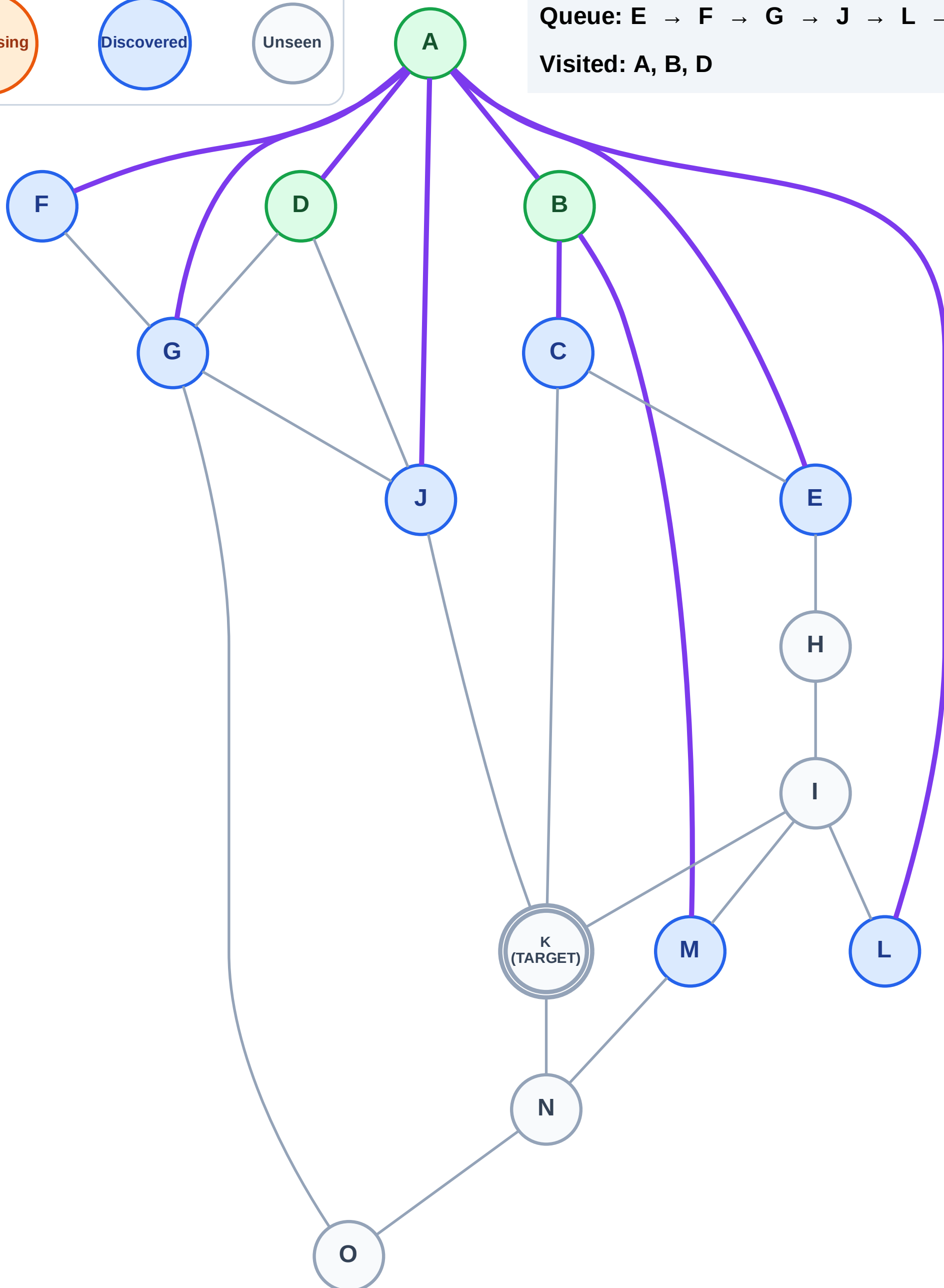
Step 7: No new neighbors from D

Legend



Queue: E → F → G → J → L → C → M

Visited: A, B, D



BFS Traversal

Step 8: Process node E

Legend

Complete

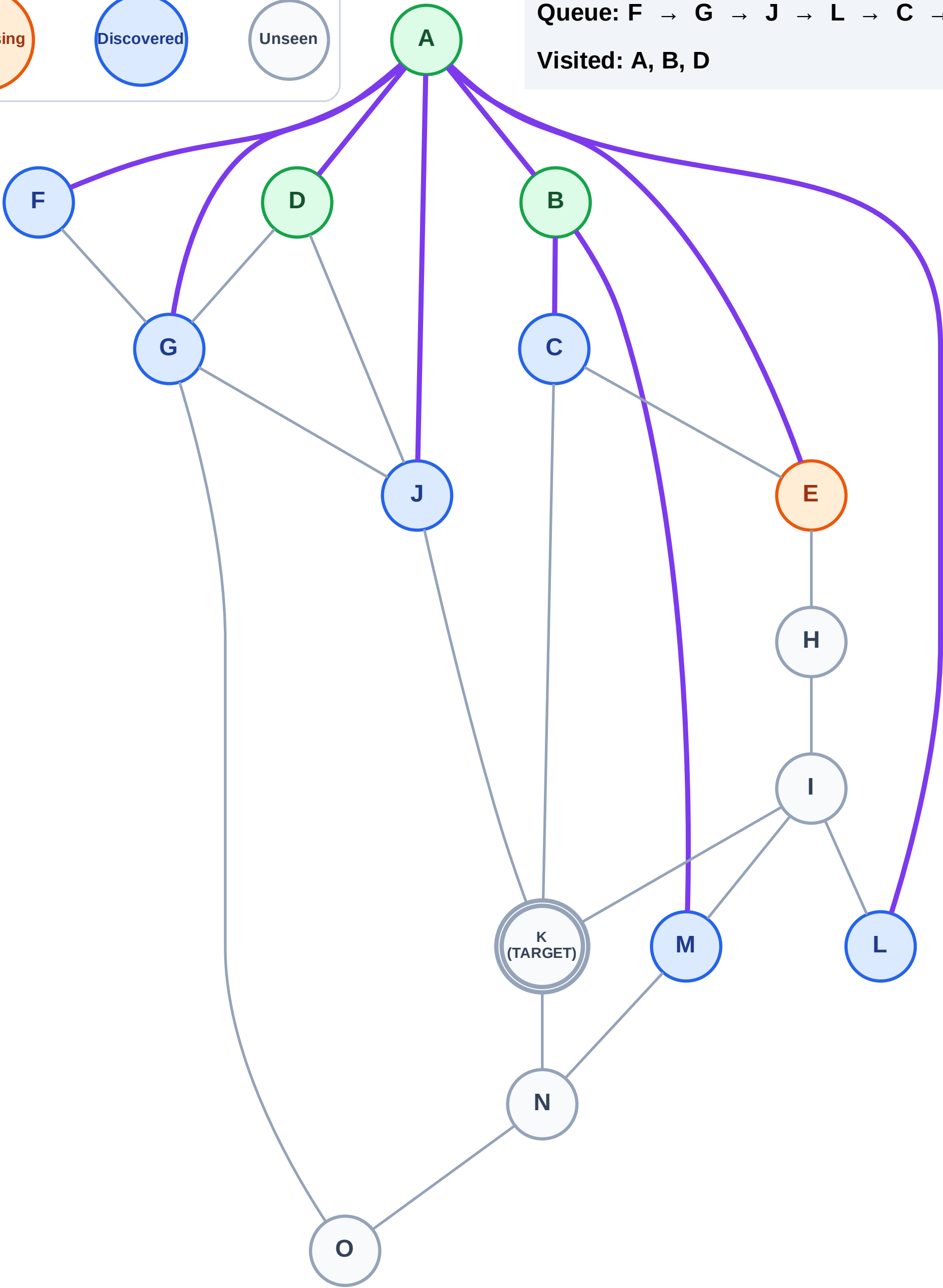
Processing

Discovered

Unseen

Queue: F → G → J → L → C → M

Visited: A, B, D



BFS Traversal

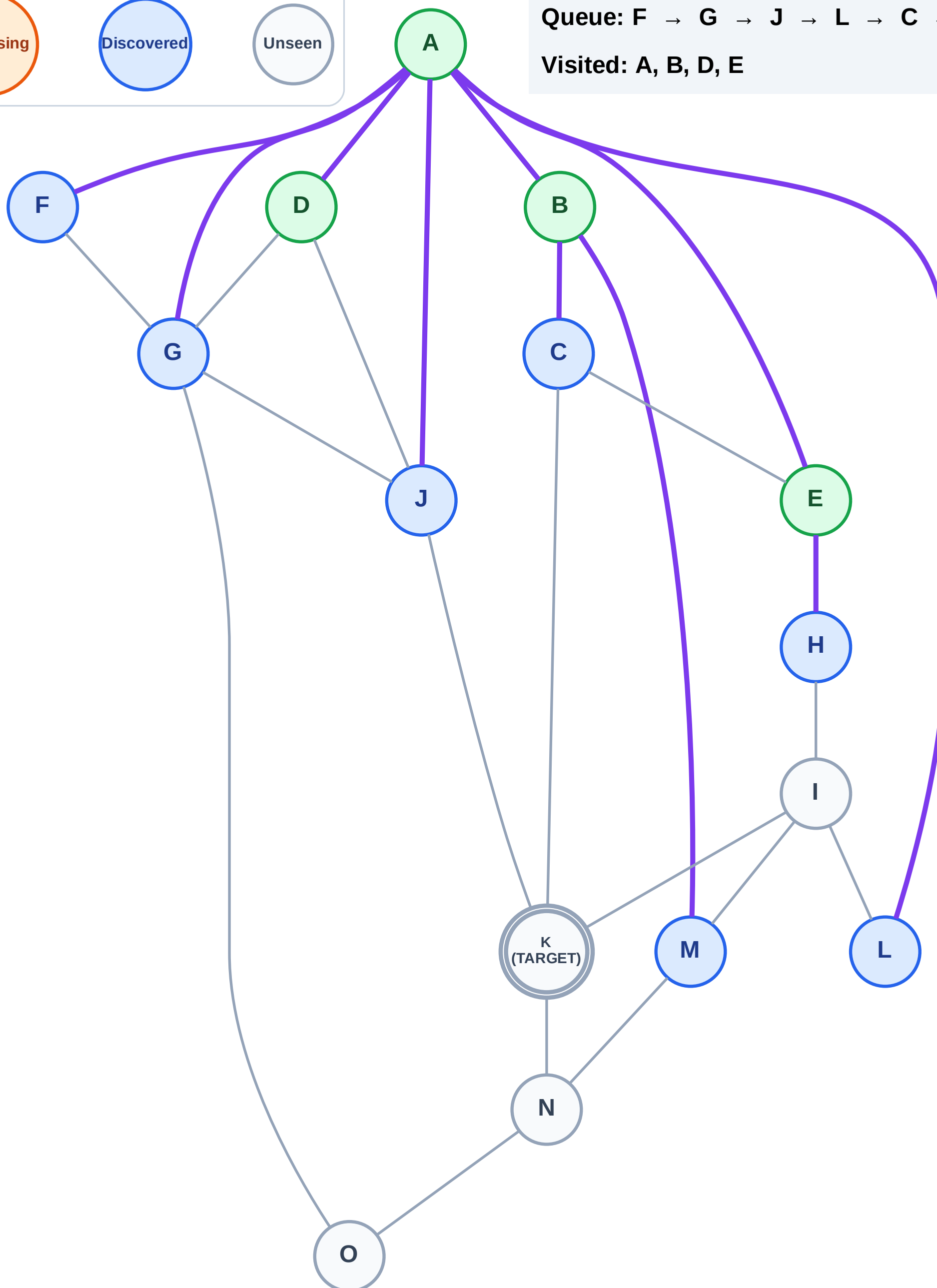
Step 9: Discover H from E

Legend



Queue: F → G → J → L → C → M → H

Visited: A, B, D, E



BFS Traversal

Step 10: Process node F

Legend

Complete

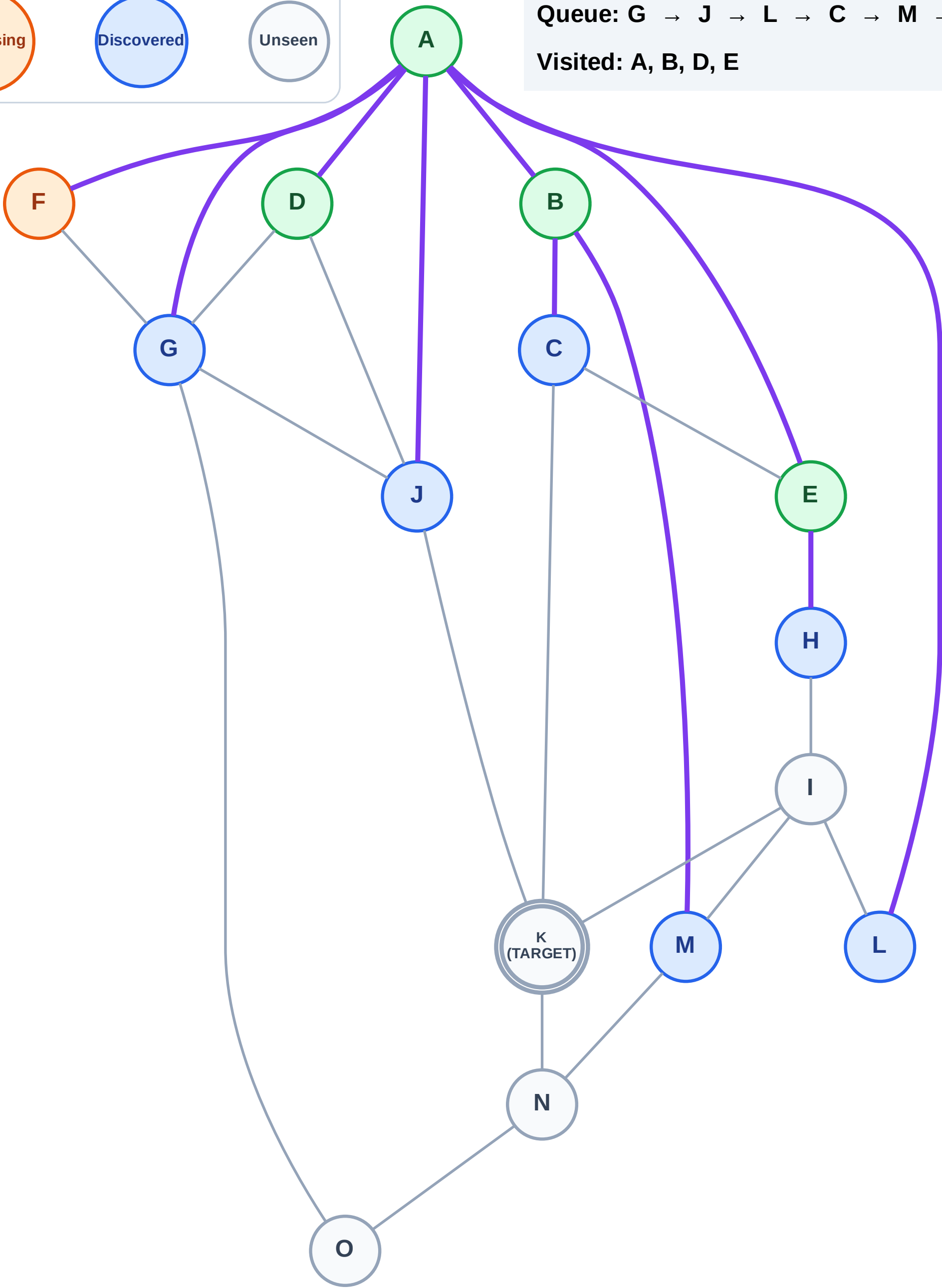
Processing

Discovered

Unseen

Queue: G → J → L → C → M → H

Visited: A, B, D, E

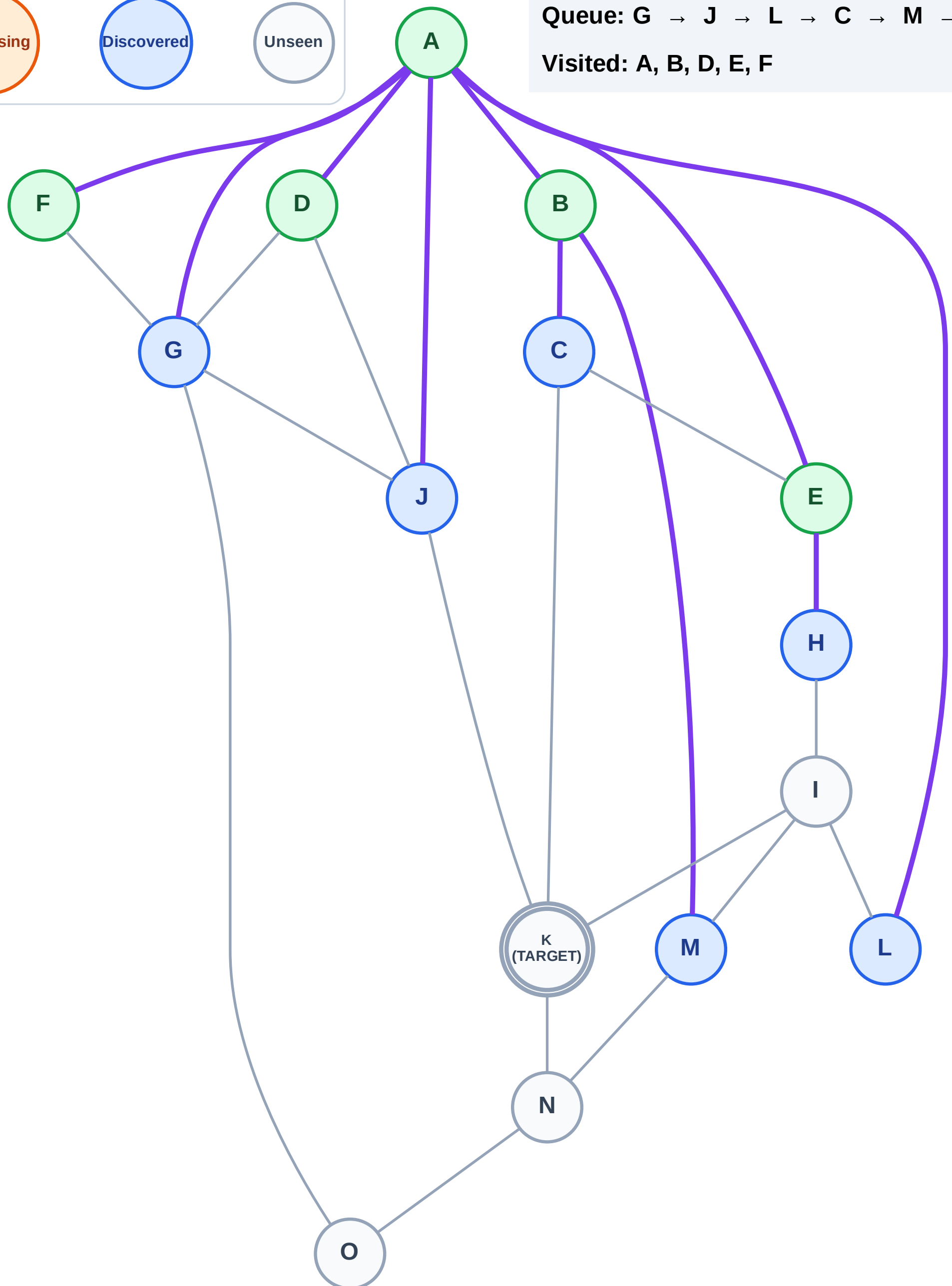


Step 11: No new neighbors from F

Legend

Unseen

Visited: A, B, D, E, F



BFS Traversal

Step 12: Process node G

Legend

Complete

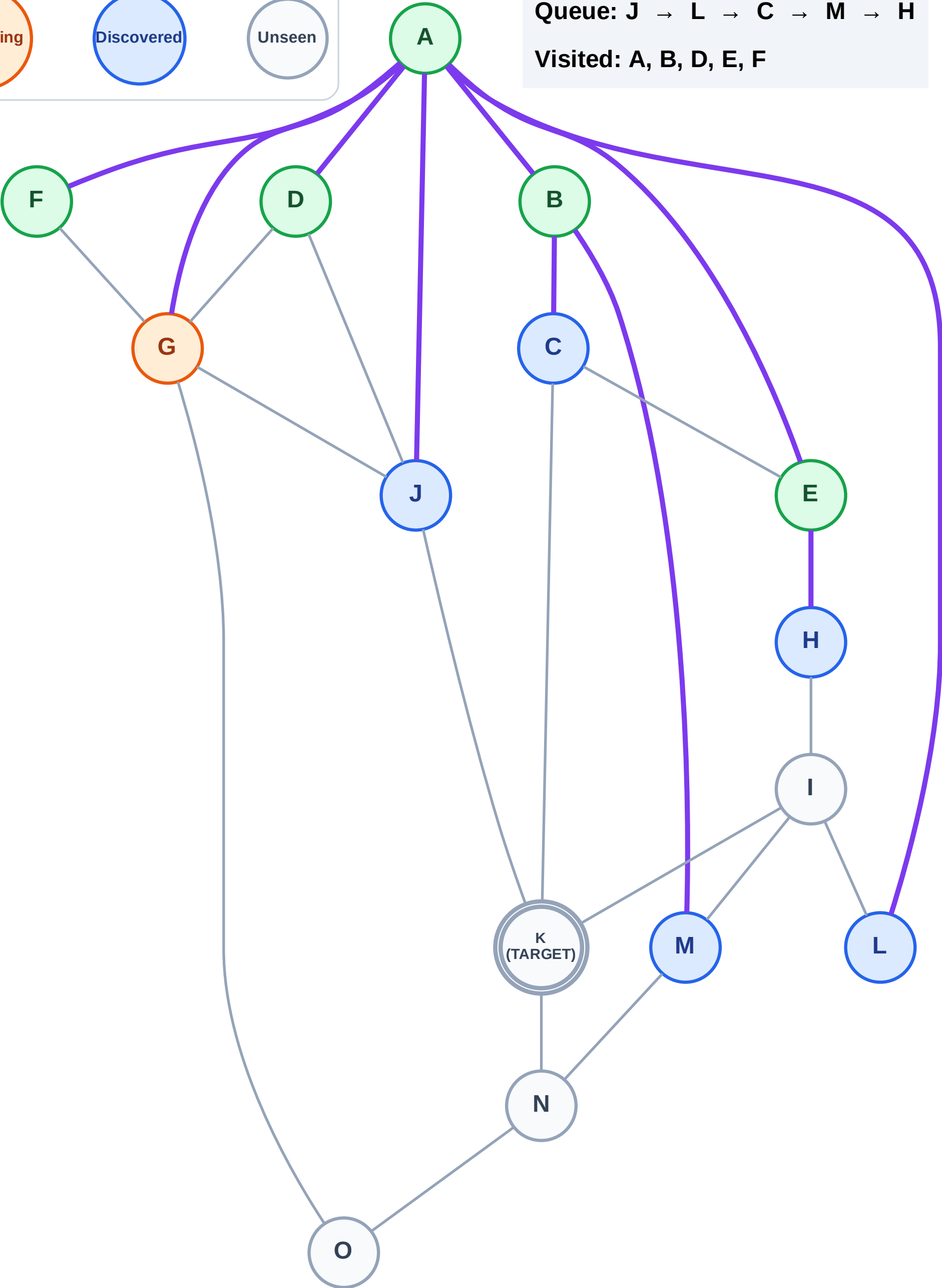
Processing

Discovered

Unseen

Queue: J → L → C → M → H

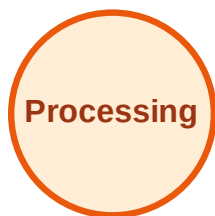
Visited: A, B, D, E, F



BFS Traversal

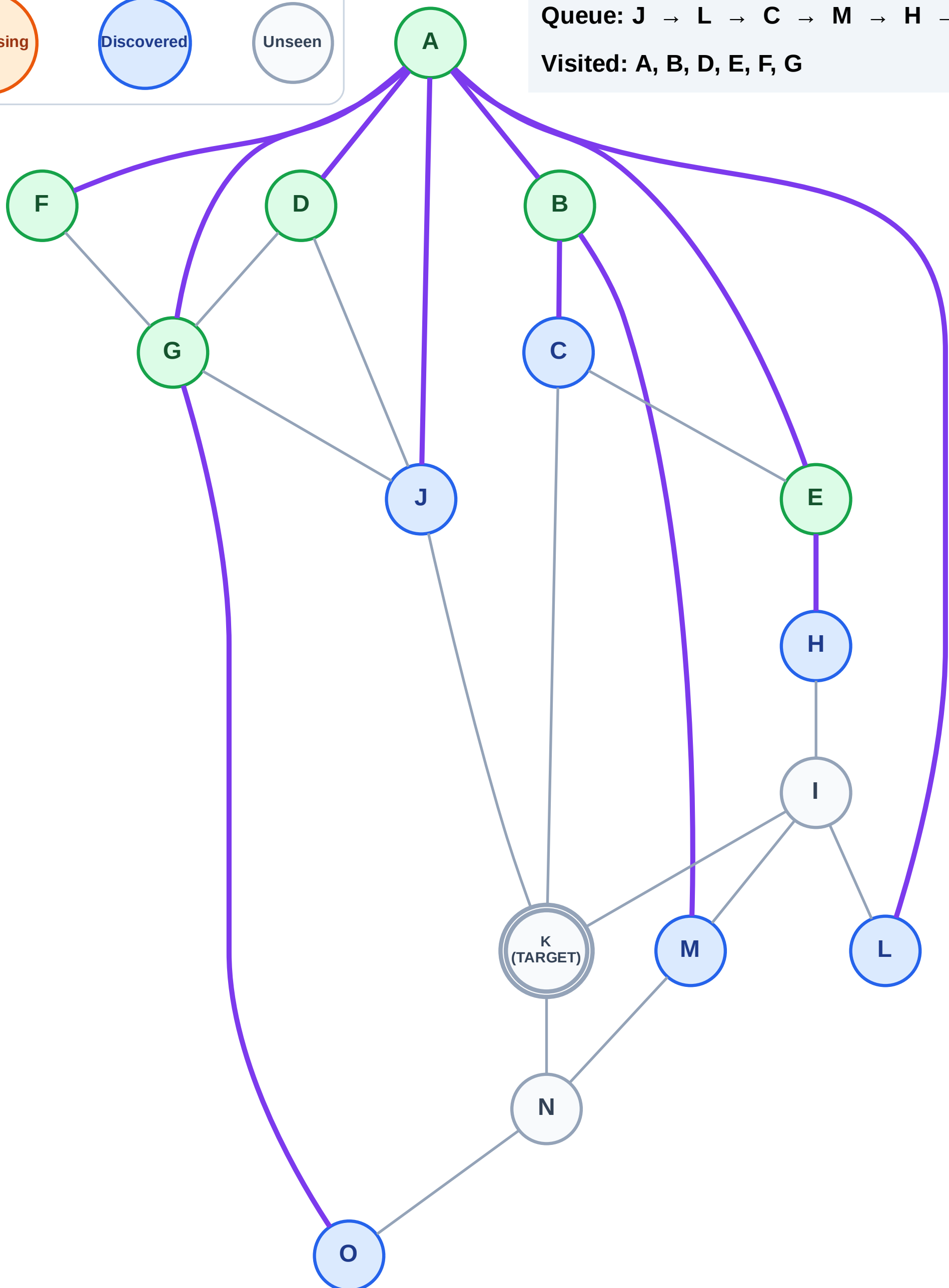
Step 13: Discover O from G

Legend



Queue: J → L → C → M → H → O

Visited: A, B, D, E, F, G



BFS Traversal

Step 14: Process node J

Legend

Complete

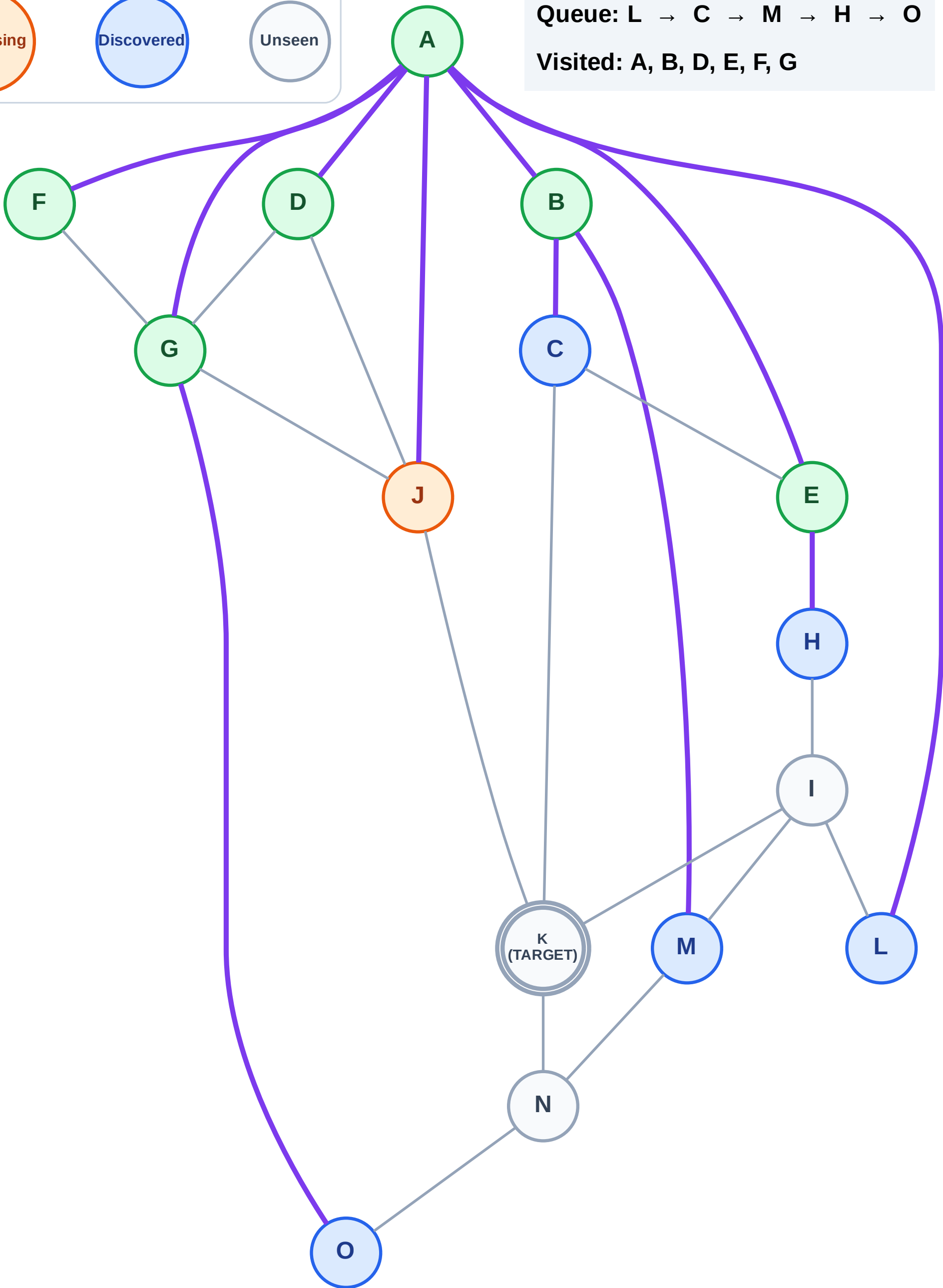
Processing

Discovered

Unseen

Queue: L → C → M → H → O

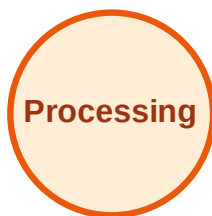
Visited: A, B, D, E, F, G



BFS Traversal

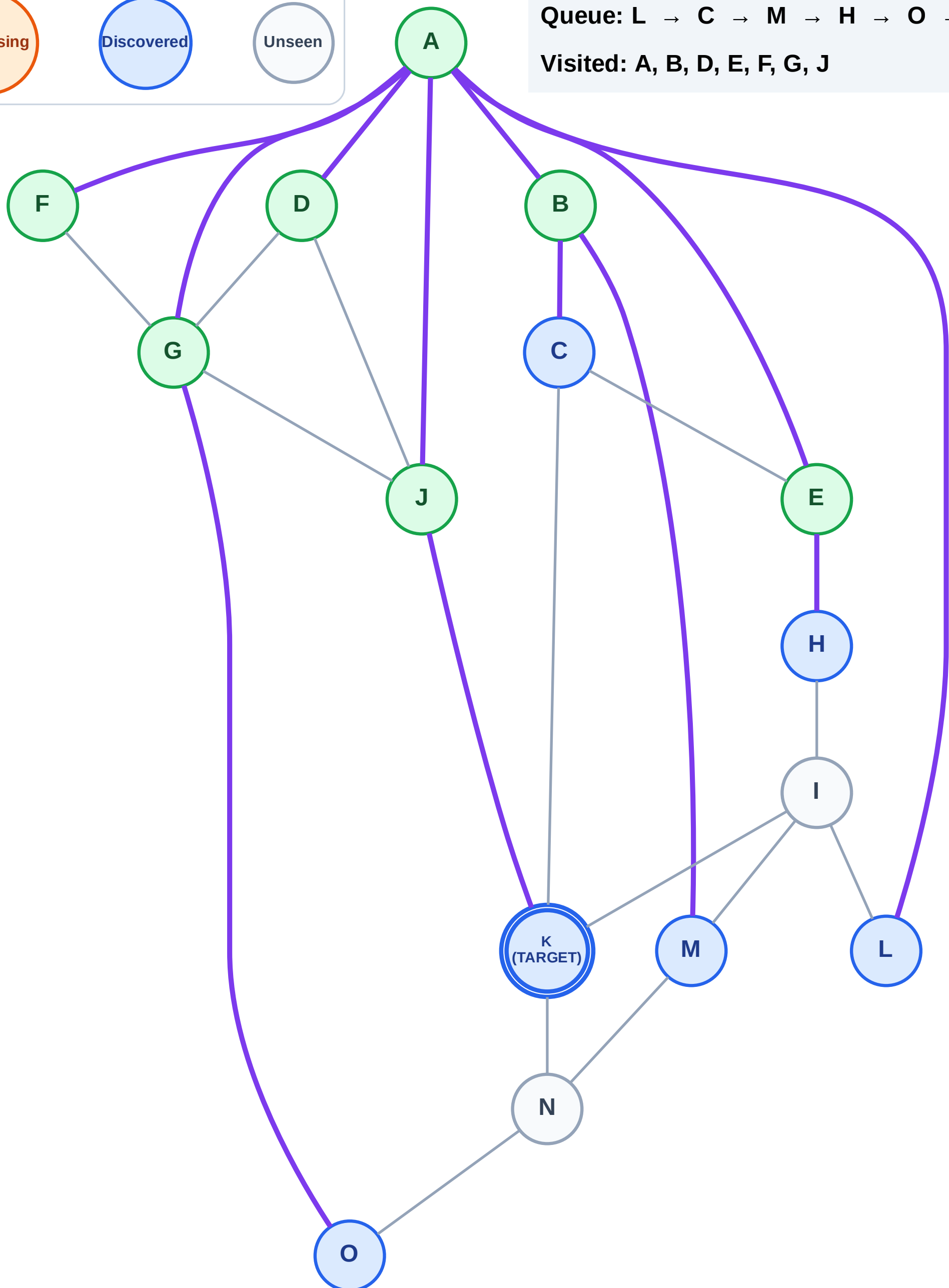
Step 15: Discover K from J

Legend



Queue: L → C → M → H → O → K

Visited: A, B, D, E, F, G, J



BFS Traversal

Step 16: Process node L

Legend

Complete

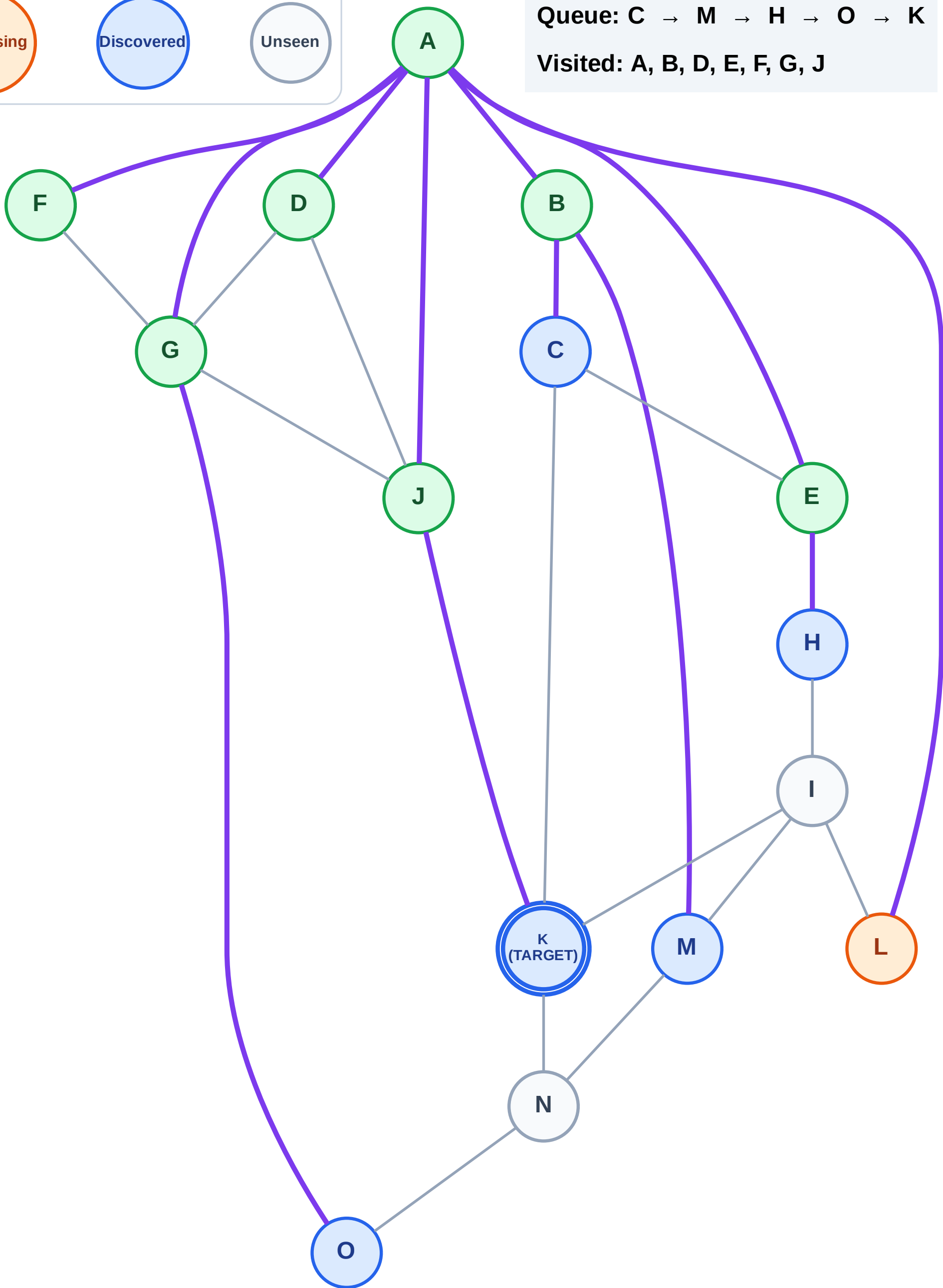
Processing

Discovered

Unseen

Queue: C → M → H → O → K

Visited: A, B, D, E, F, G, J

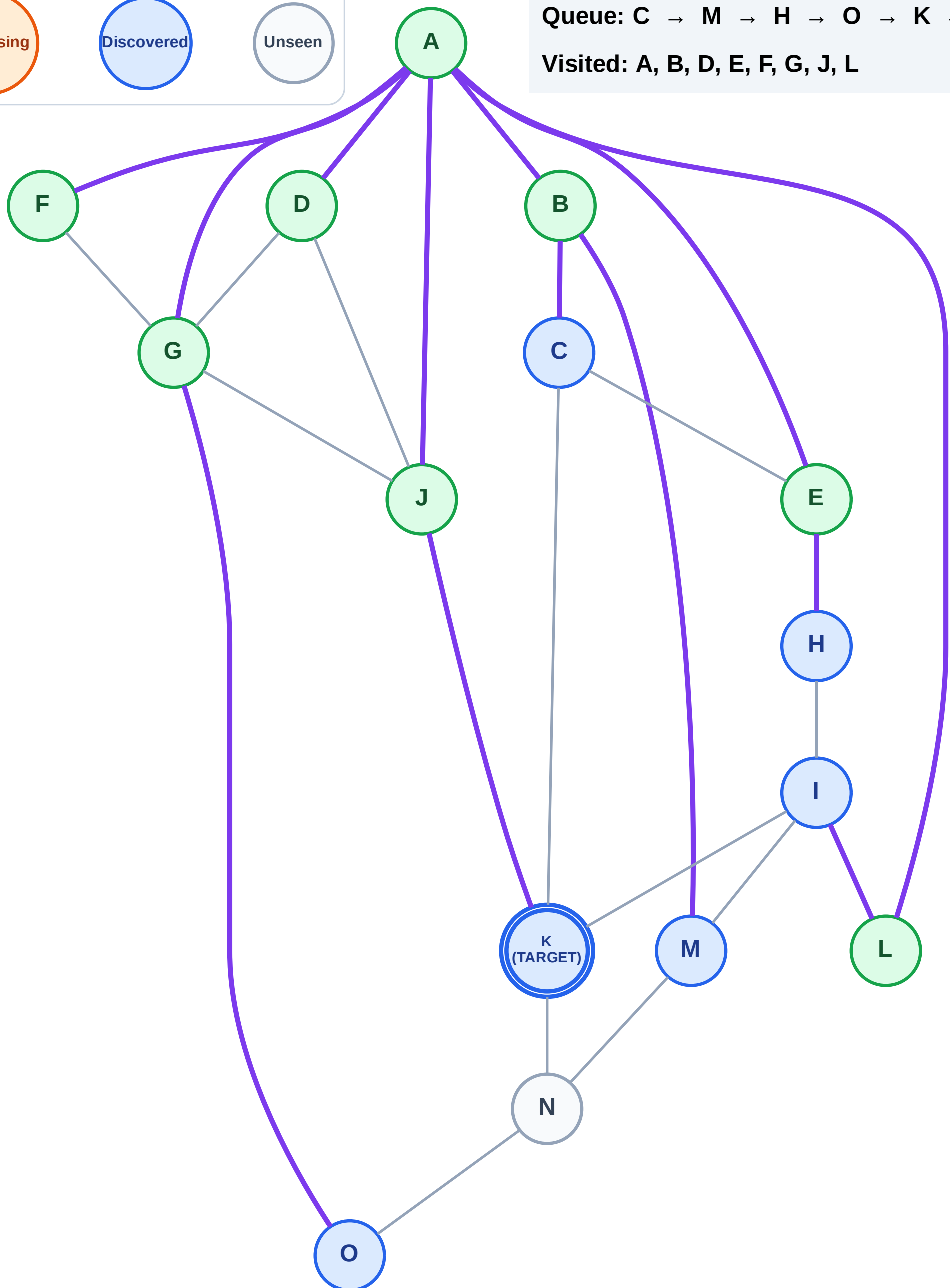


Step 17: Discover I from L

Legend

Unseen

Visited: A, B, D, E, F, G, J, L



BFS Traversal

Step 18: Process node C

Legend

Complete

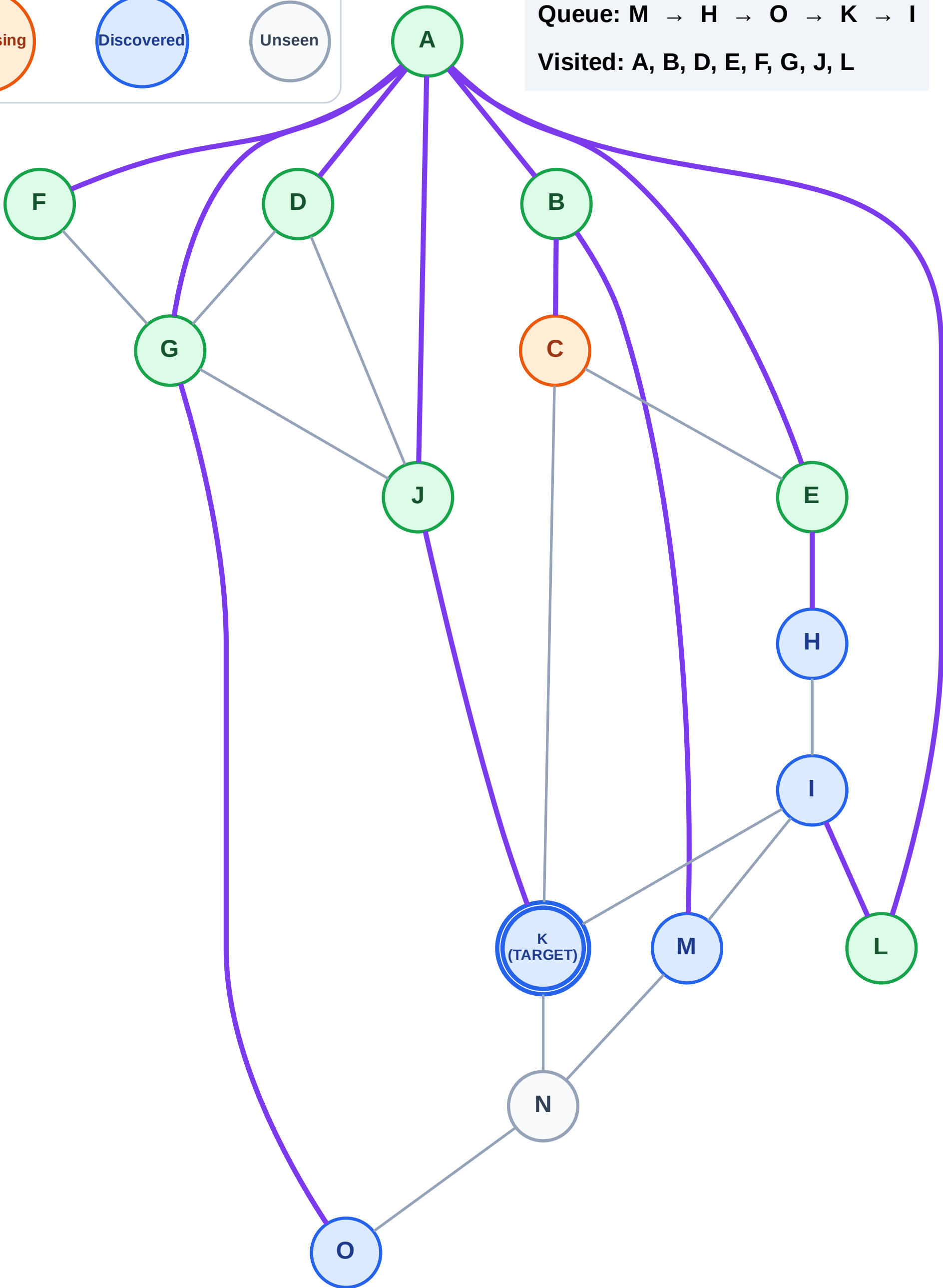
Processing

Discovered

Unseen

Queue: M → H → O → K → I

Visited: A, B, D, E, F, G, J, L



BFS Traversal

Step 19: No new neighbors from C

Legend

Complete

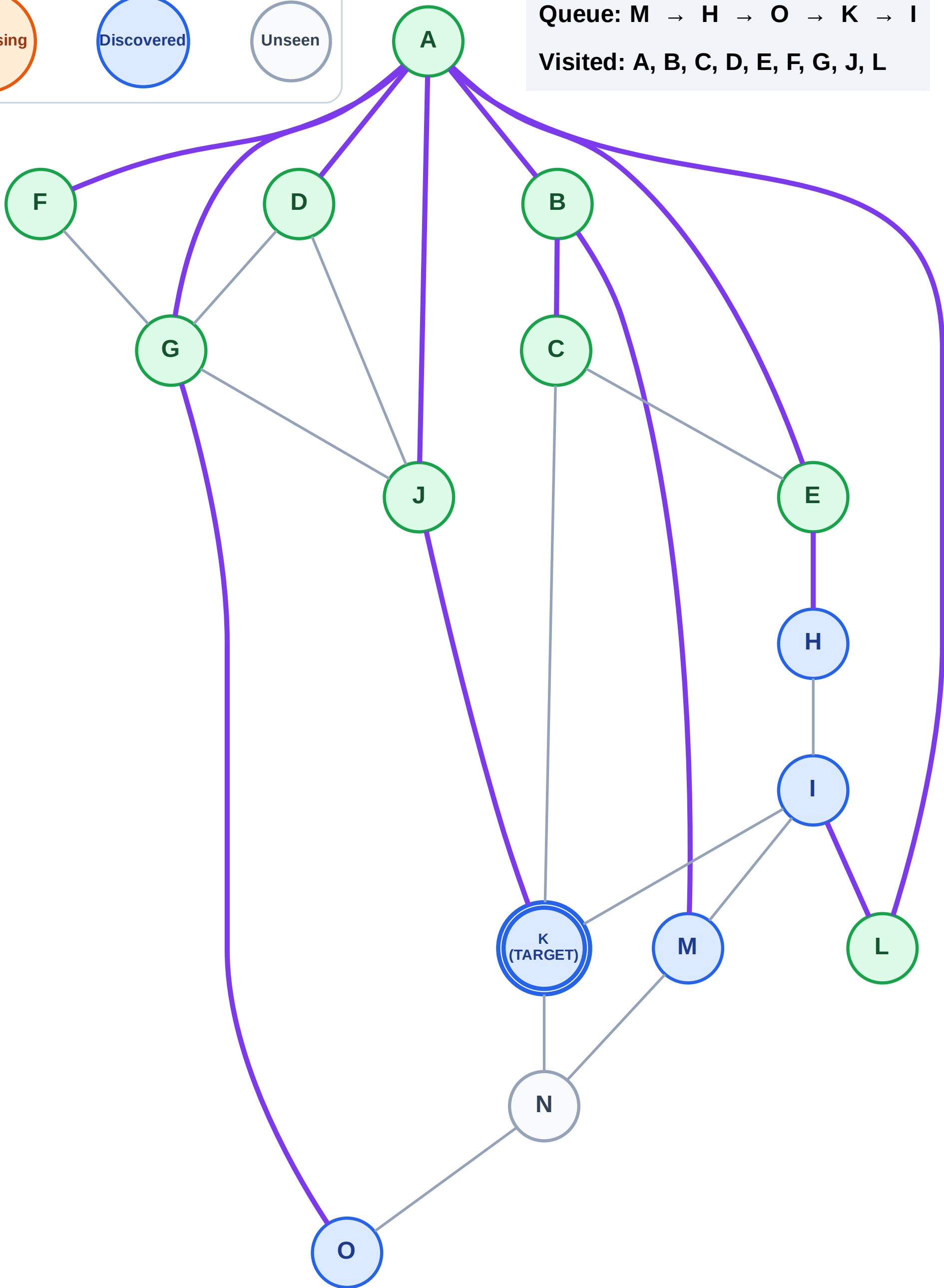
Processing

Discovered

Unseen

Queue: M → H → O → K → I

Visited: A, B, C, D, E, F, G, J, L



BFS Traversal

Step 20: Process node M

Legend

Complete

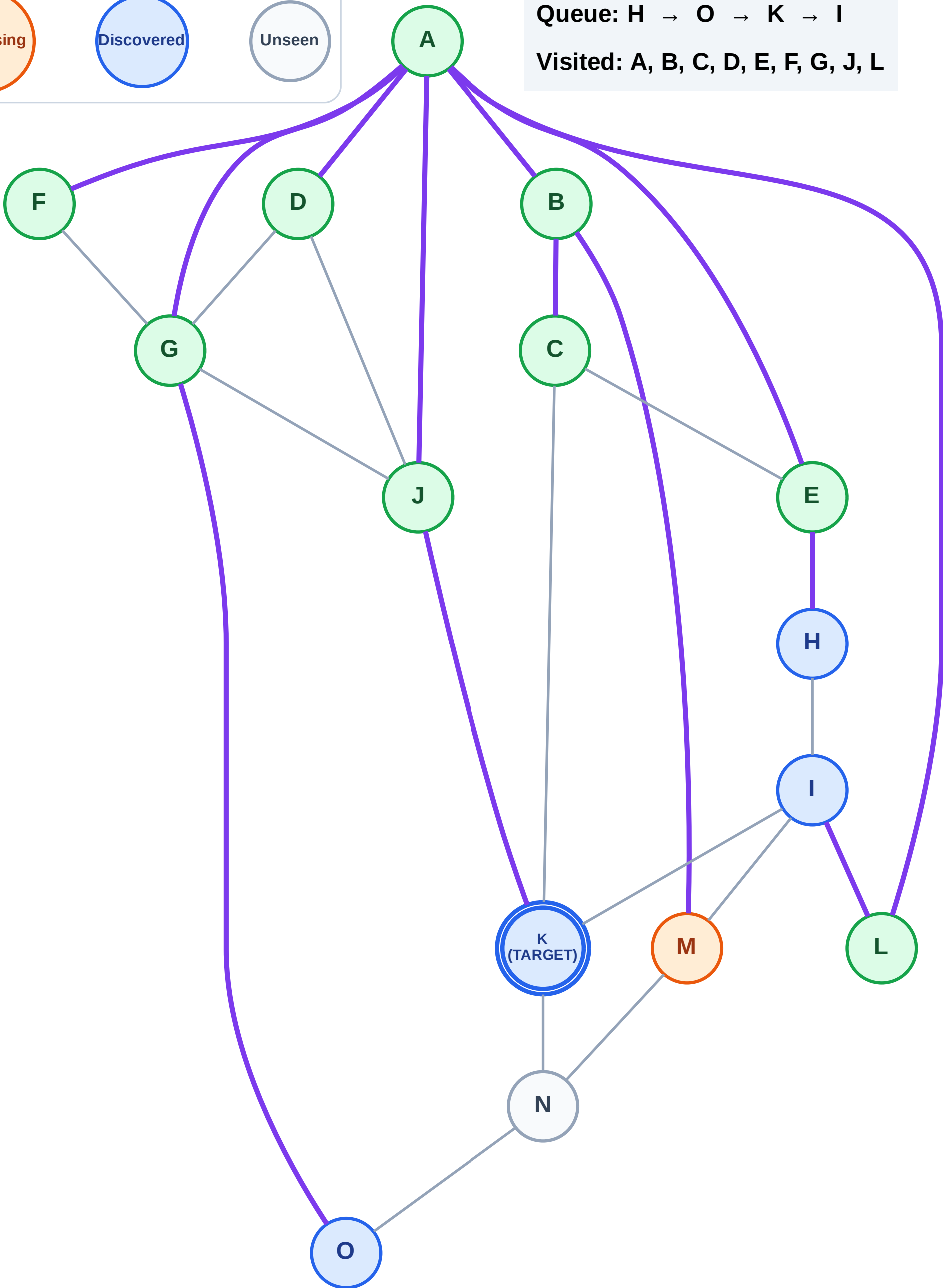
Processing

Discovered

Unseen

Queue: H → O → K → I

Visited: A, B, C, D, E, F, G, J, L



BFS Traversal

Step 21: Discover N from M

Legend

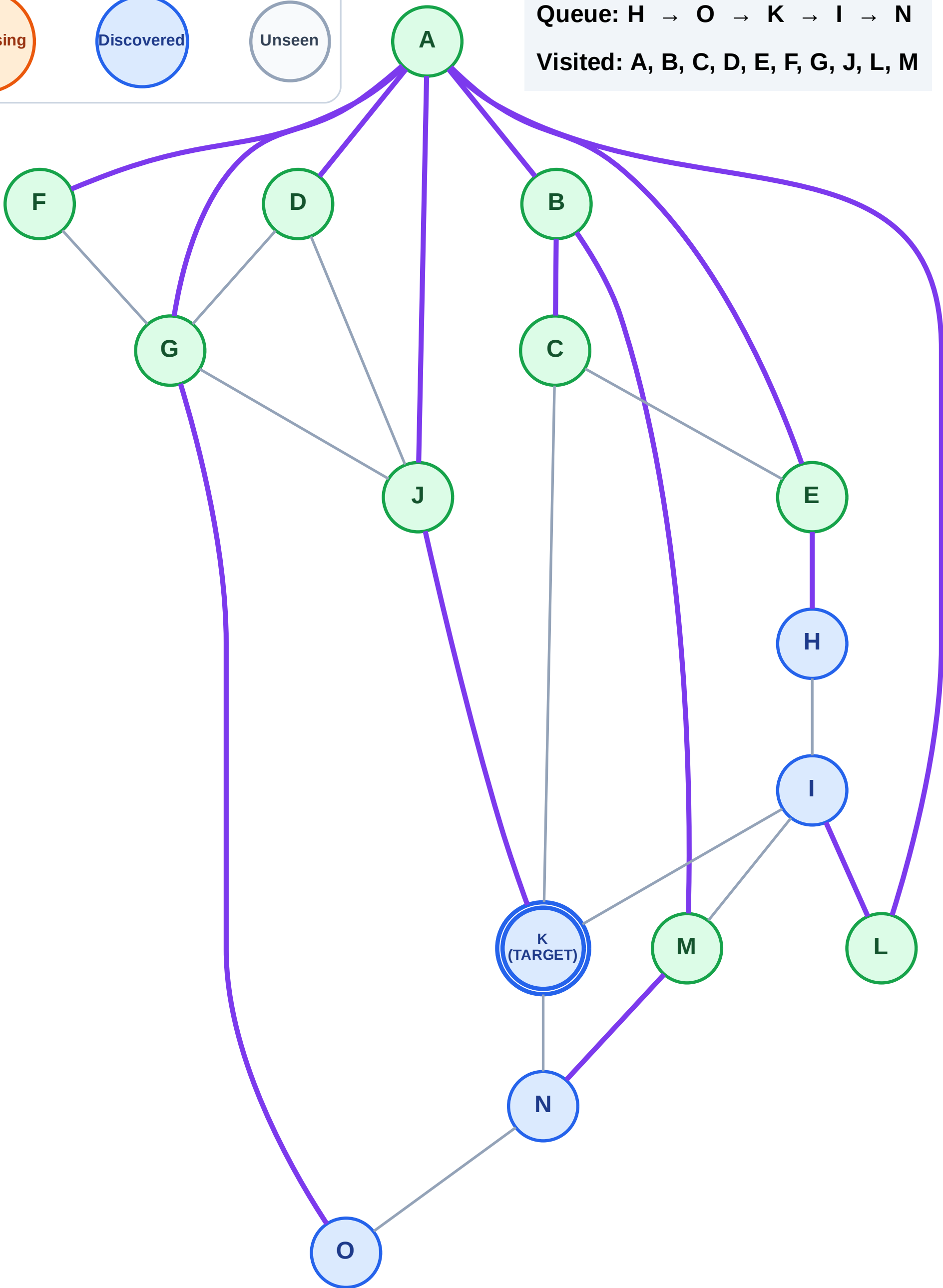
Complete

Processing

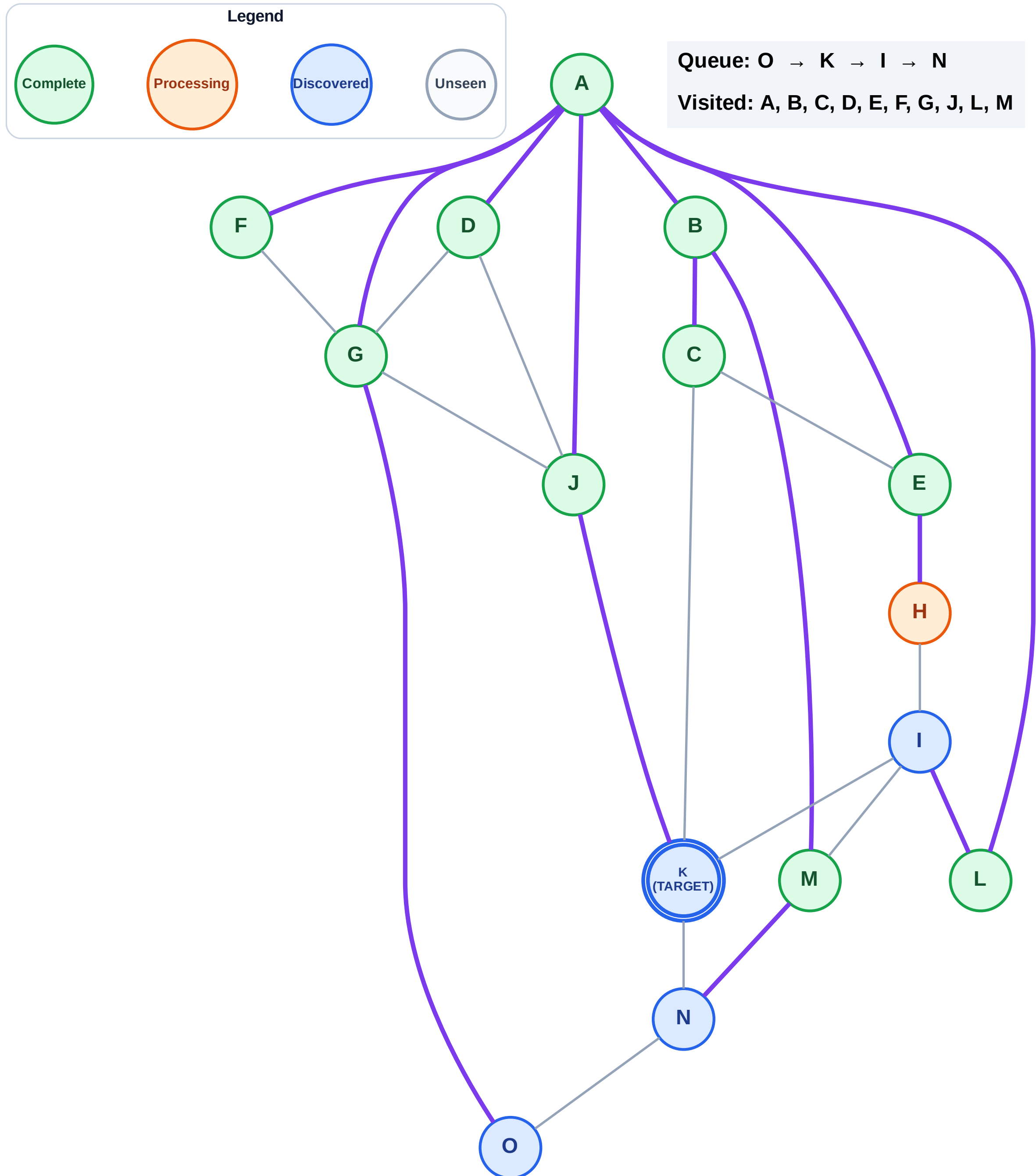
Discovered

Unseen

Queue: H → O → K → I → N
Visited: A, B, C, D, E, F, G, J, L, M



Step 22: Process node H



BFS Traversal

Step 23: No new neighbors from H

Legend

Complete

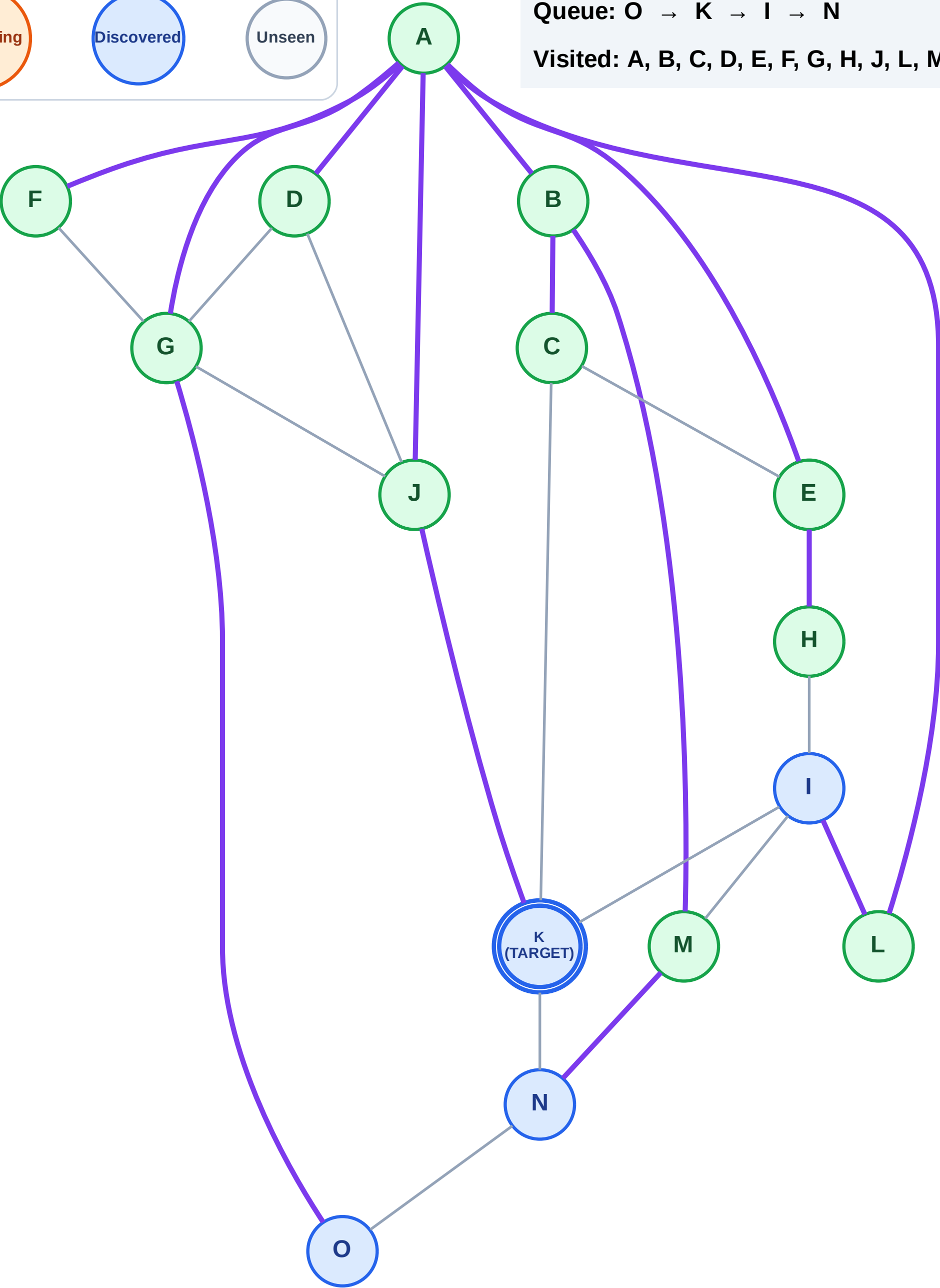
Processing

Discovered

Unseen

Queue: O → K → I → N

Visited: A, B, C, D, E, F, G, H, J, L, M



BFS Traversal

Step 24: Process node O

Legend

Complete

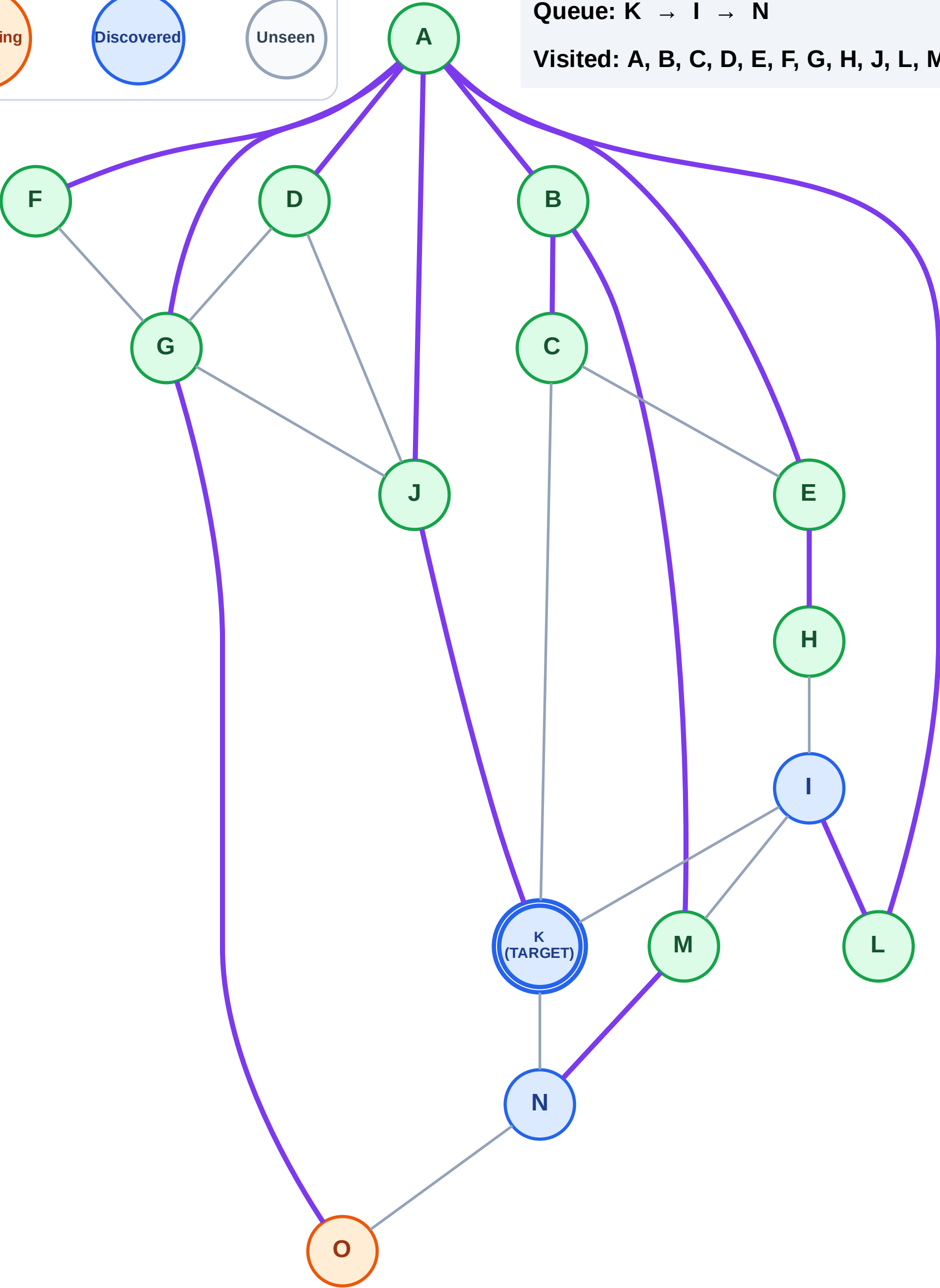
Processing

Discovered

Unseen

Queue: K → I → N

Visited: A, B, C, D, E, F, G, H, J, L, M



BFS Traversal

Step 25: No new neighbors from O

Legend

Complete

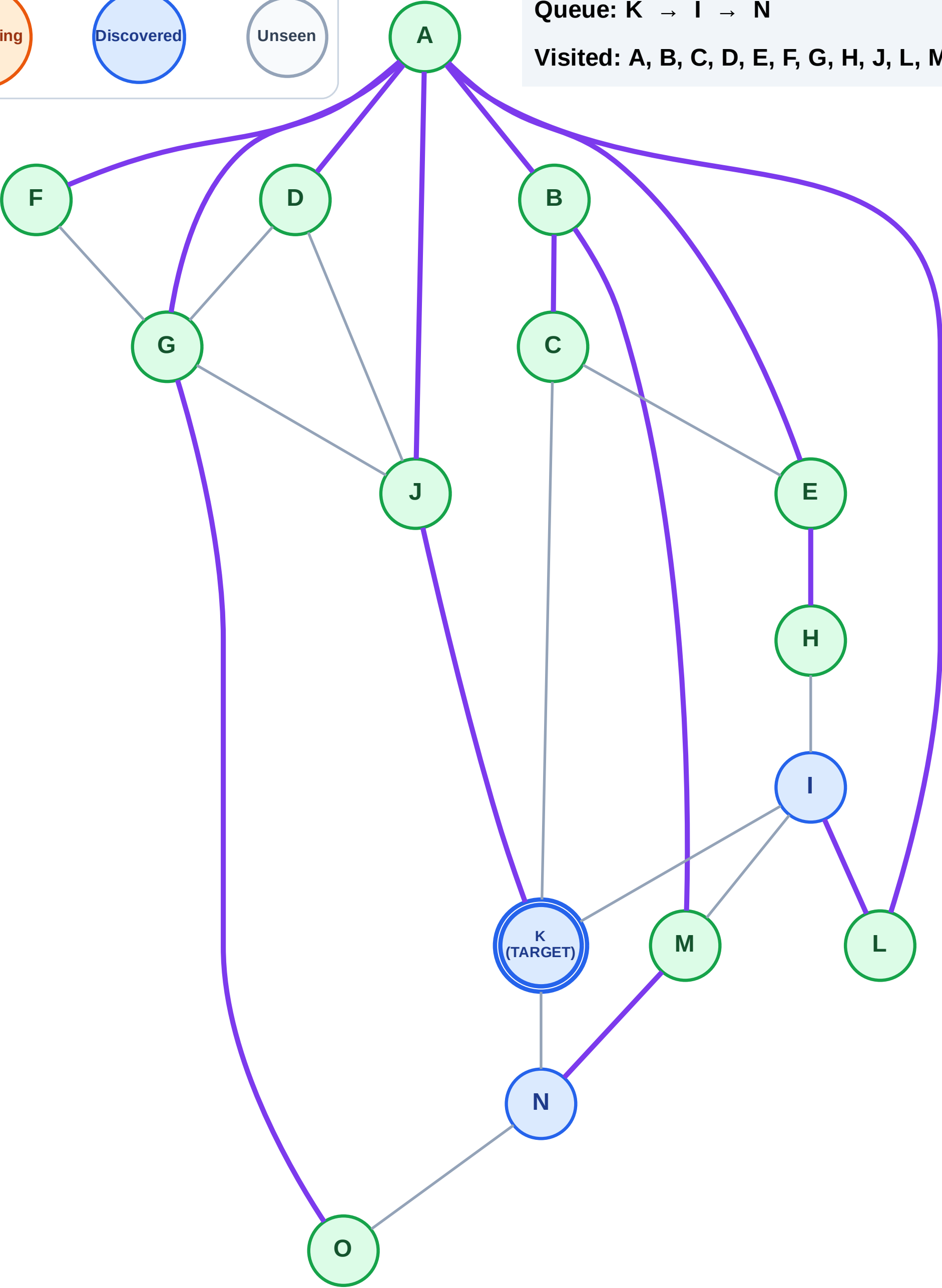
Processing

Discovered

Unseen

Queue: K → I → N

Visited: A, B, C, D, E, F, G, H, J, L, M, O



BFS Traversal

Step 26: Process node K

Legend

Complete

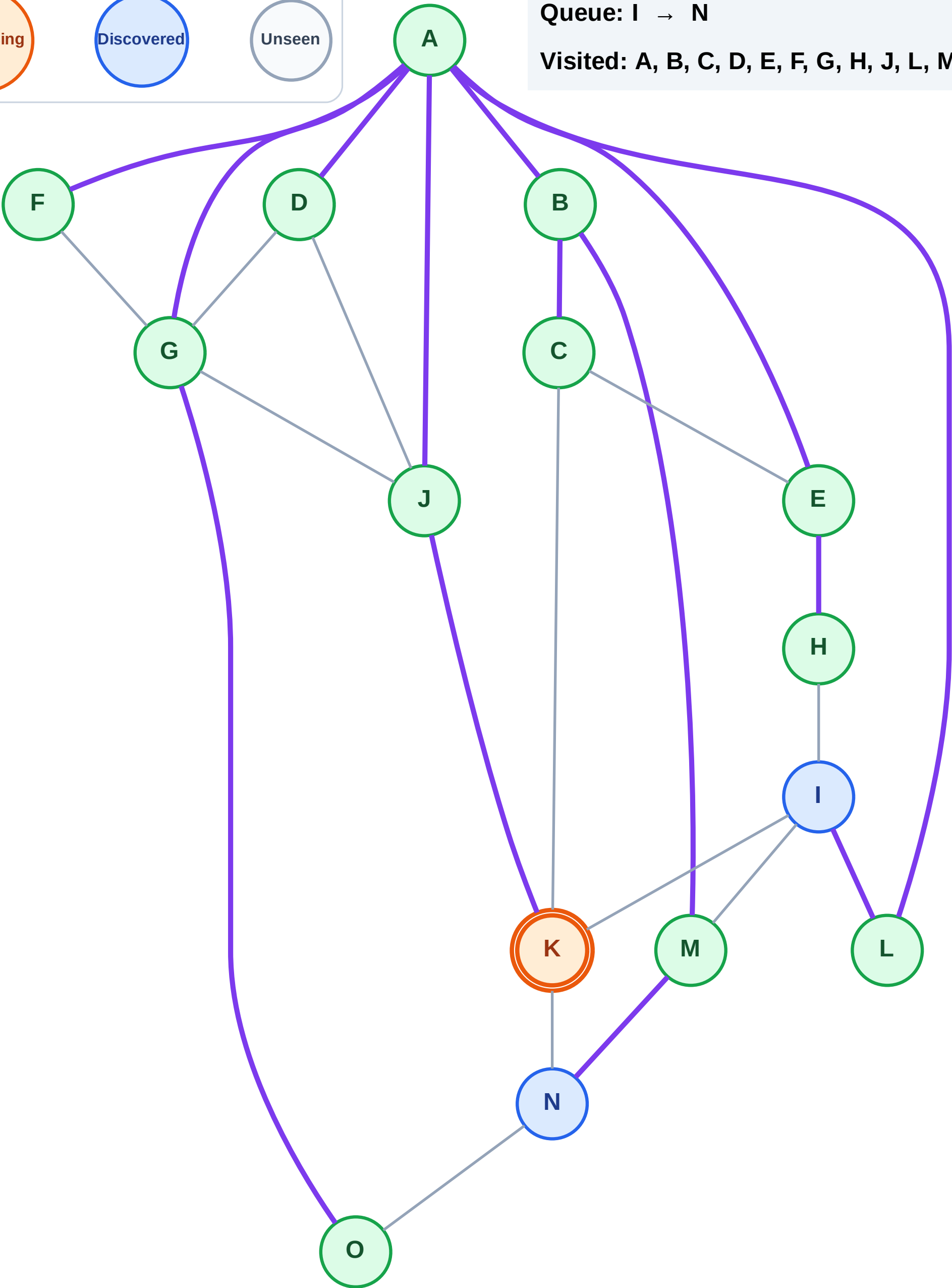
Processing

Discovered

Unseen

Queue: I → N

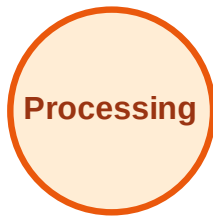
Visited: A, B, C, D, E, F, G, H, J, L, M, O



BFS Traversal

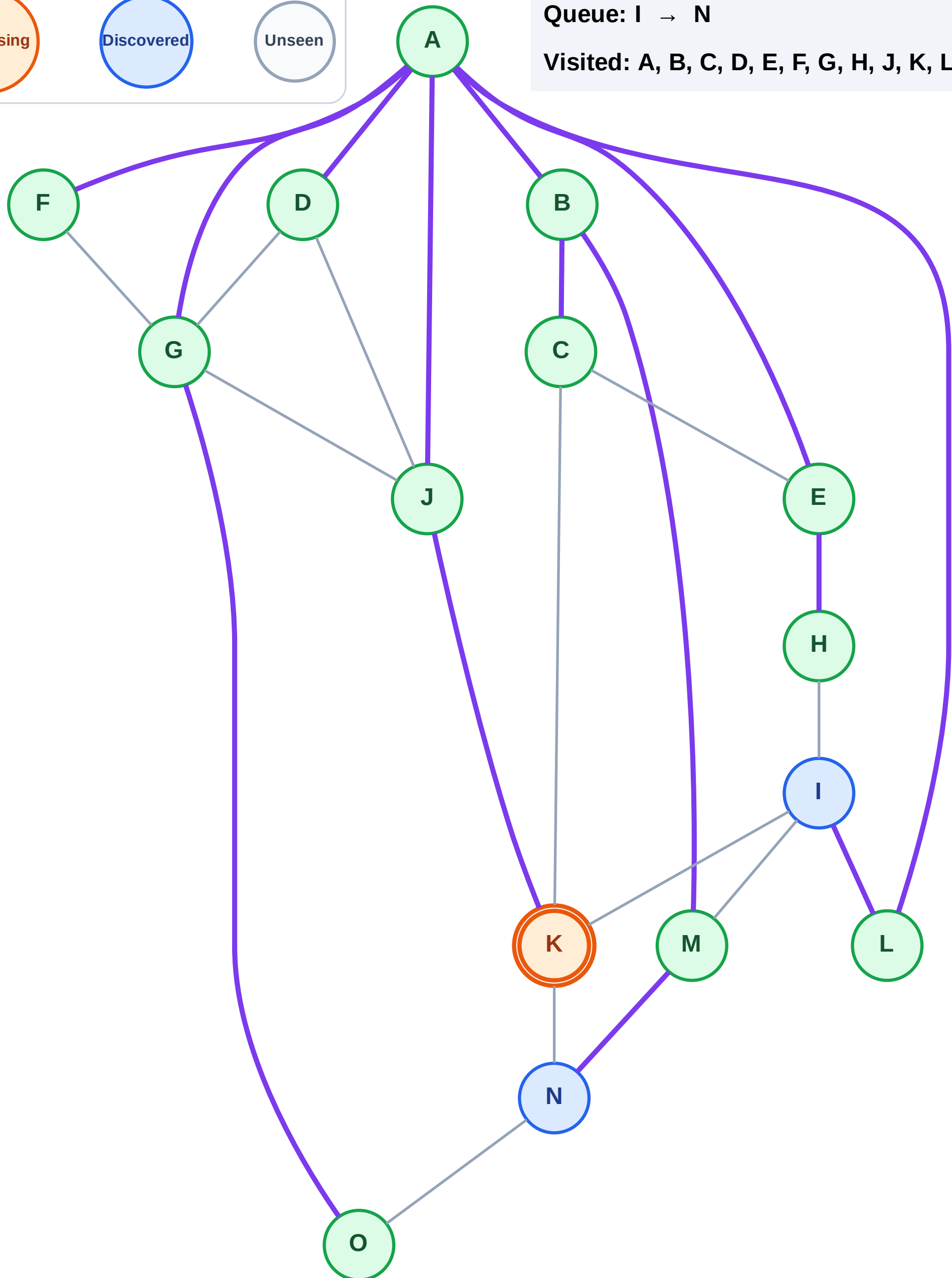
Step 27: TARGET NODE K REACHED! Path (2 hops): A → J → K

Legend



Queue: I → N

Visited: A, B, C, D, E, F, G, H, J, K, L, M, O



DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

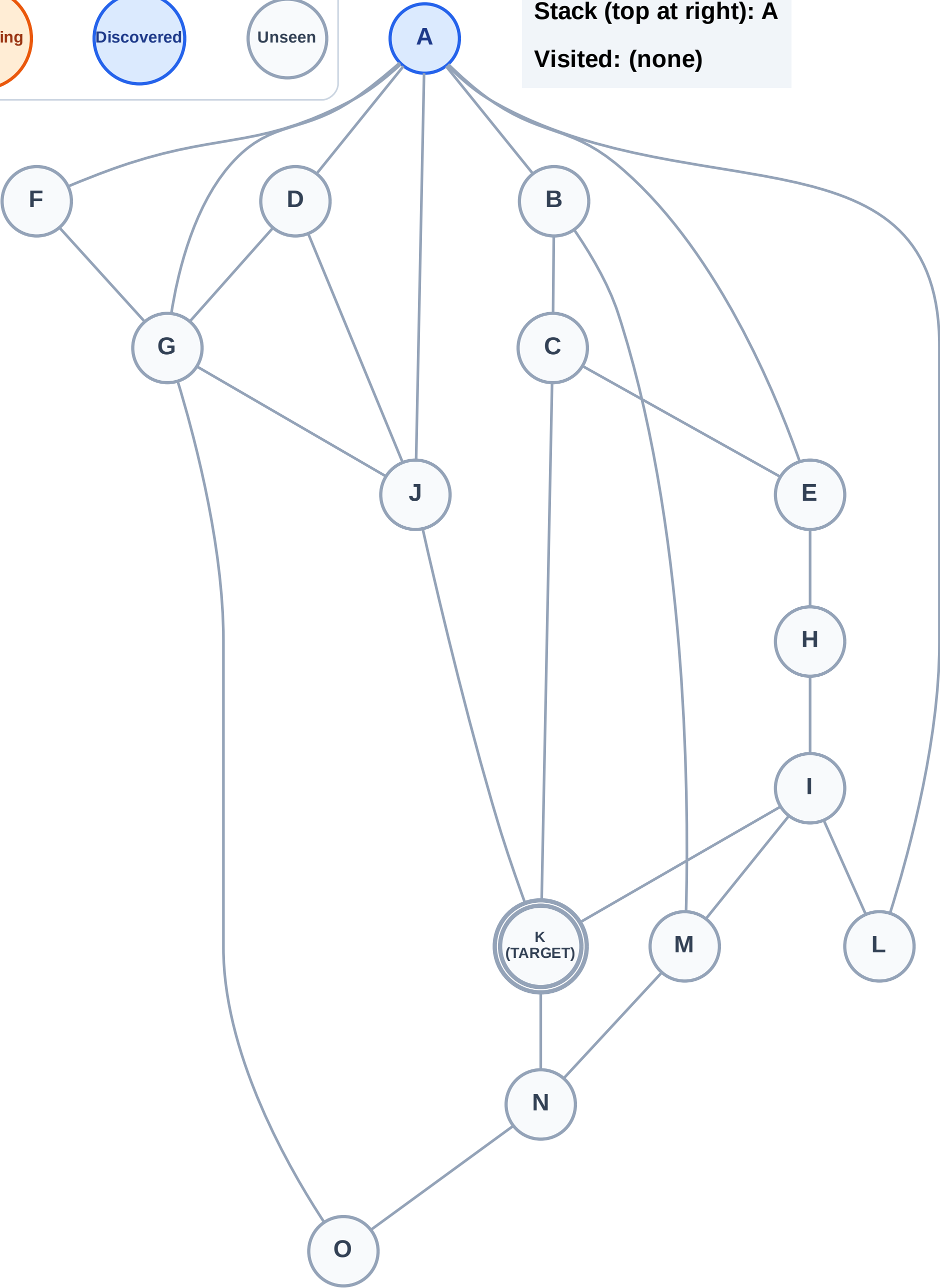
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

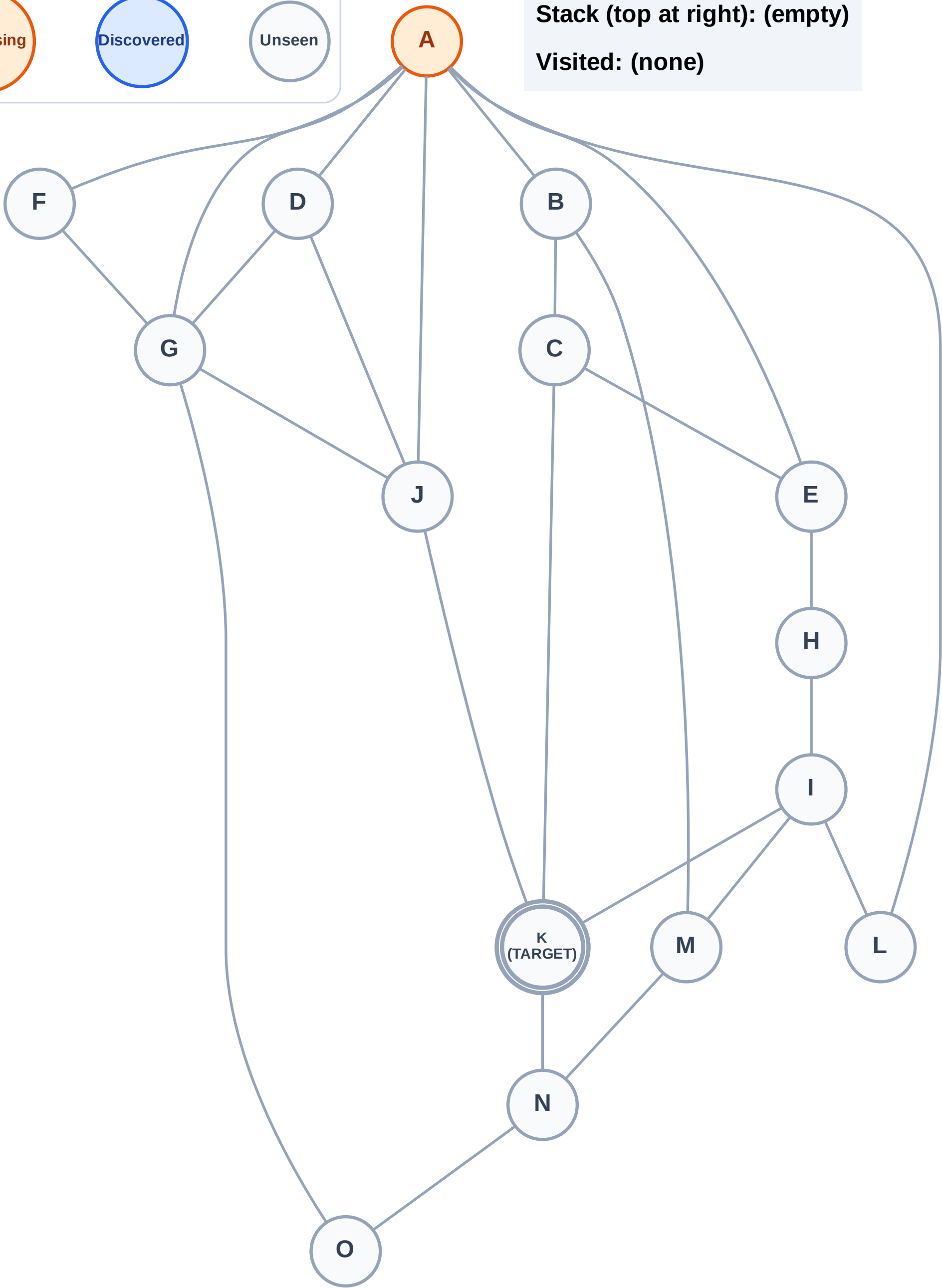
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

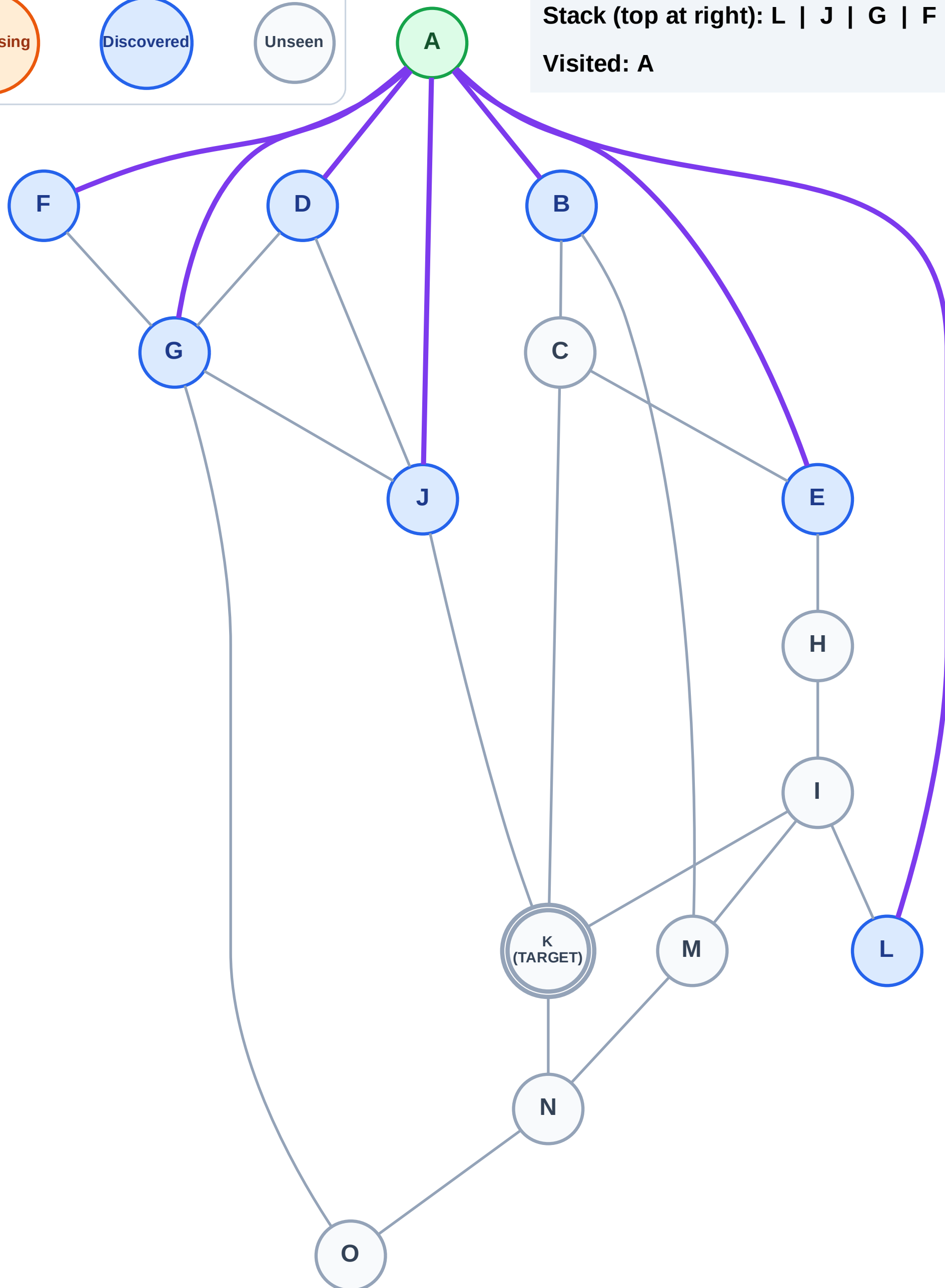
Step 3: Discover L, J, G, F, E, D, B from A

Legend



Stack (top at right): L | J | G | F | E | D | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

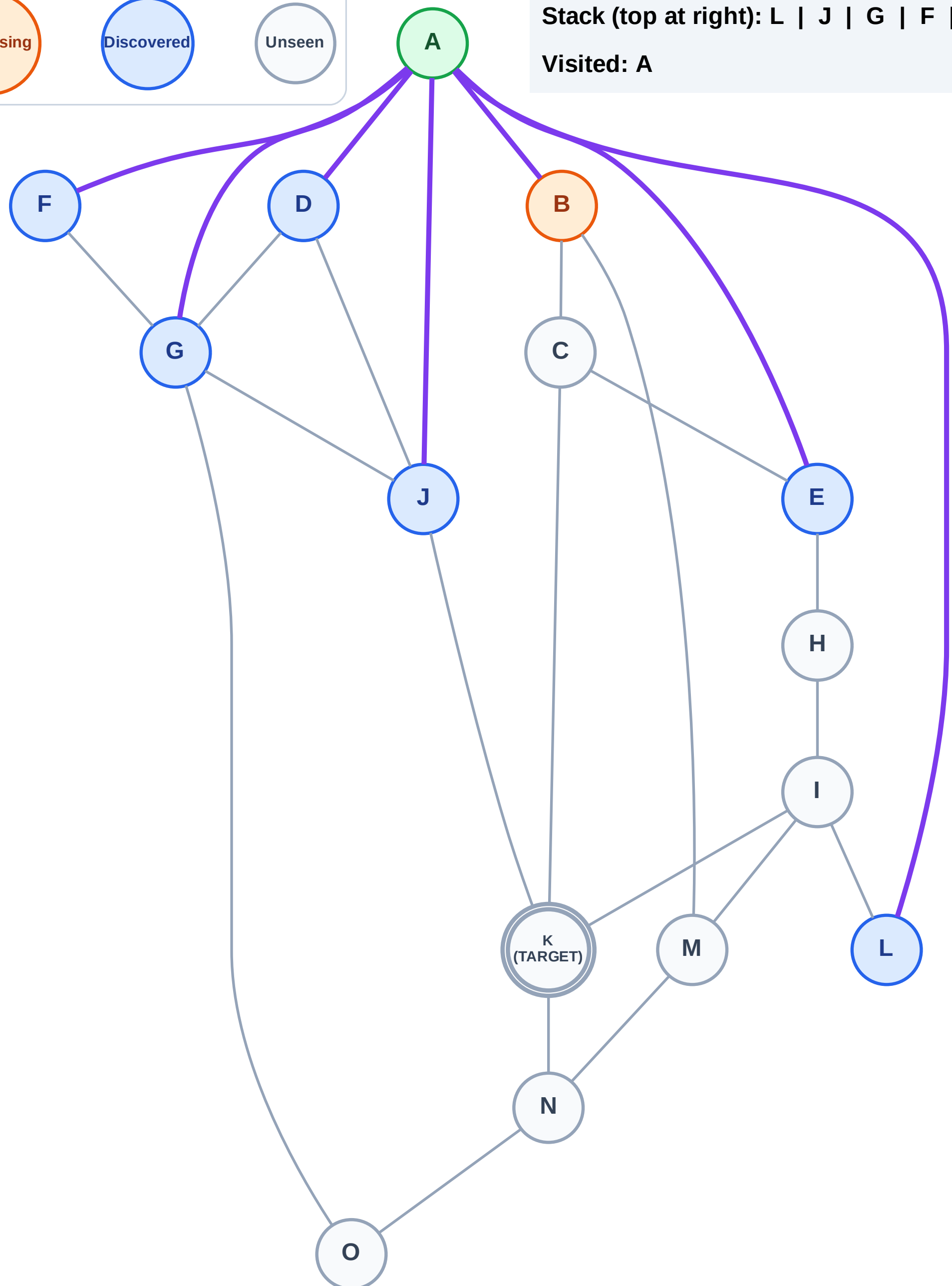
Processing

Discovered

Unseen

Stack (top at right): L | J | G | F | E | D

Visited: A



DFS Traversal

Step 5: Discover M, C from B

Legend



Complete



Processing



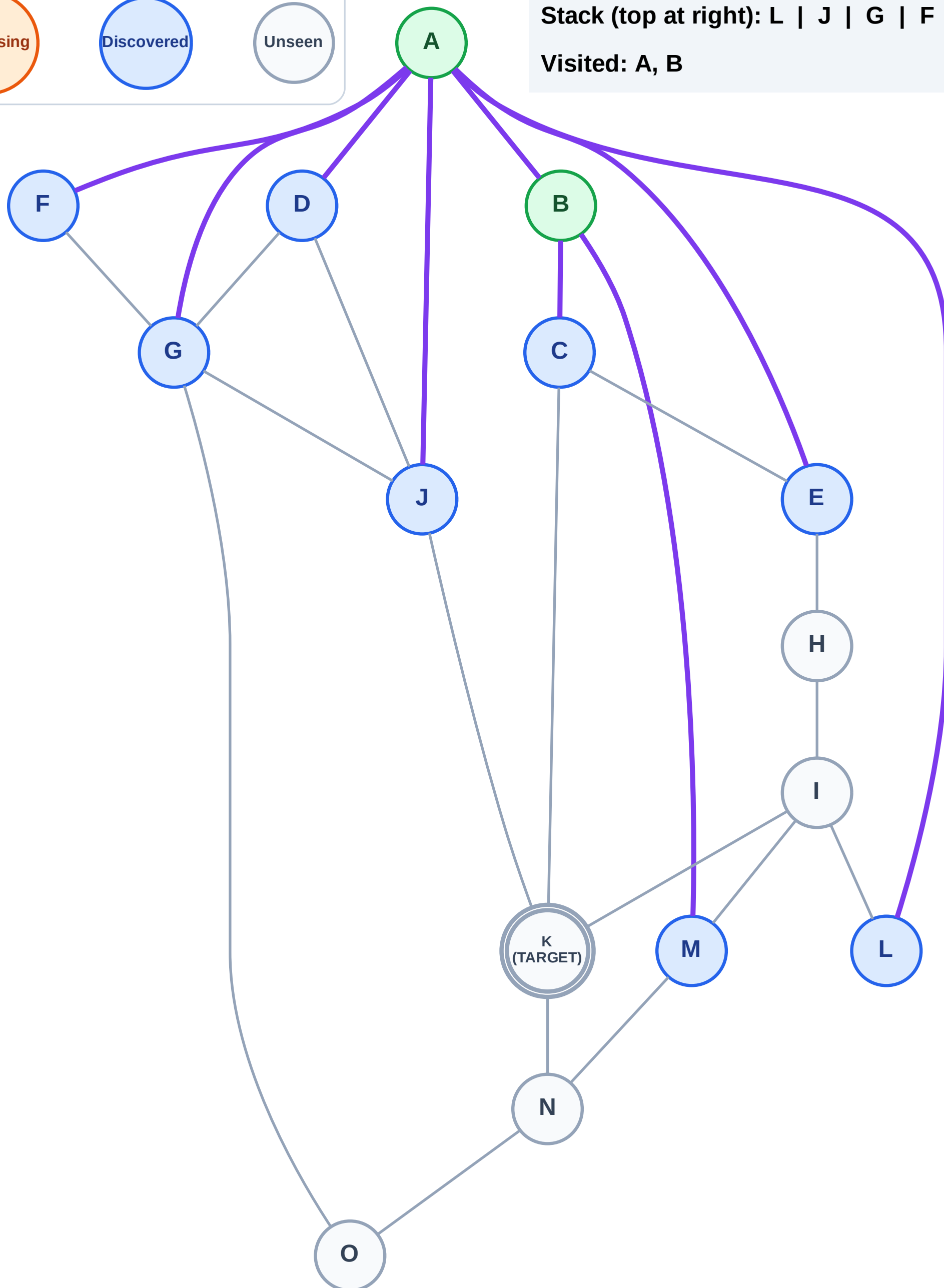
Discovered



Unseen

Stack (top at right): L | J | G | F | E | D | M | C

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

Complete

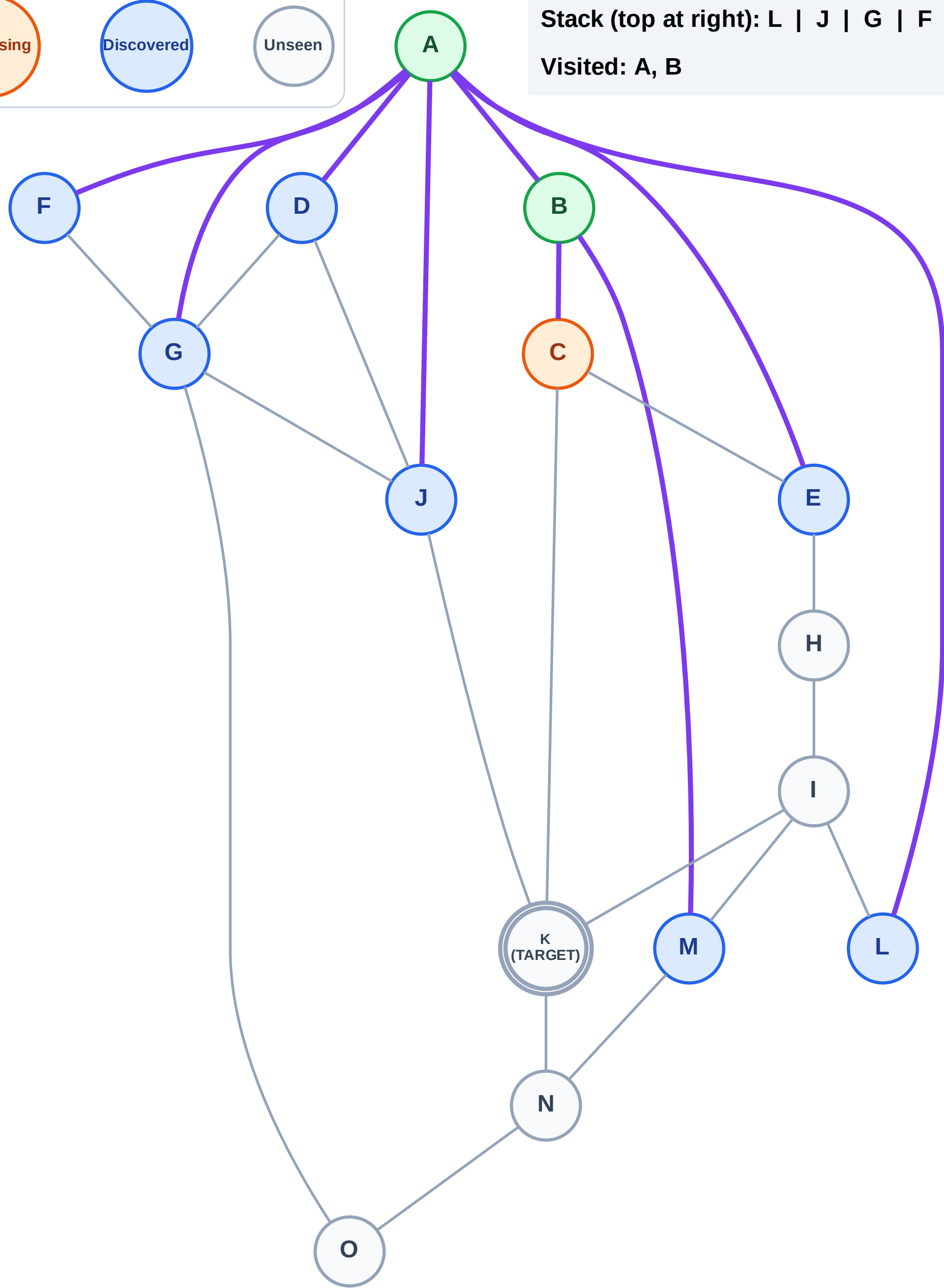
Processing

Discovered

Unseen

Stack (top at right): L | J | G | F | E | D | M

Visited: A, B



DFS Traversal

Step 7: Discover K from C

Legend



Complete



Processing



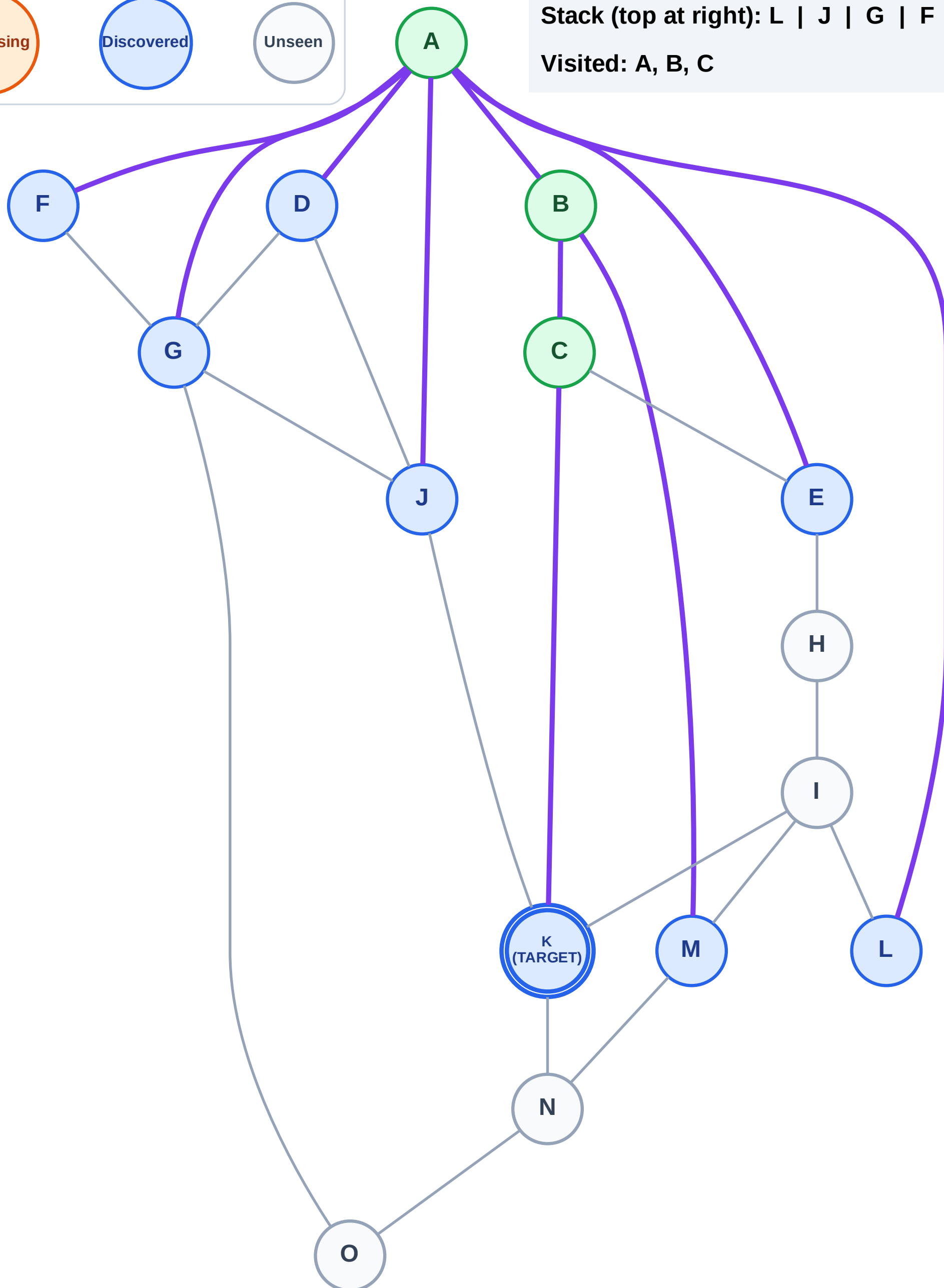
Discovered



Unseen

Stack (top at right): L | J | G | F | E | D | M | K

Visited: A, B, C



Step 8: Process node K

Legend

Complete

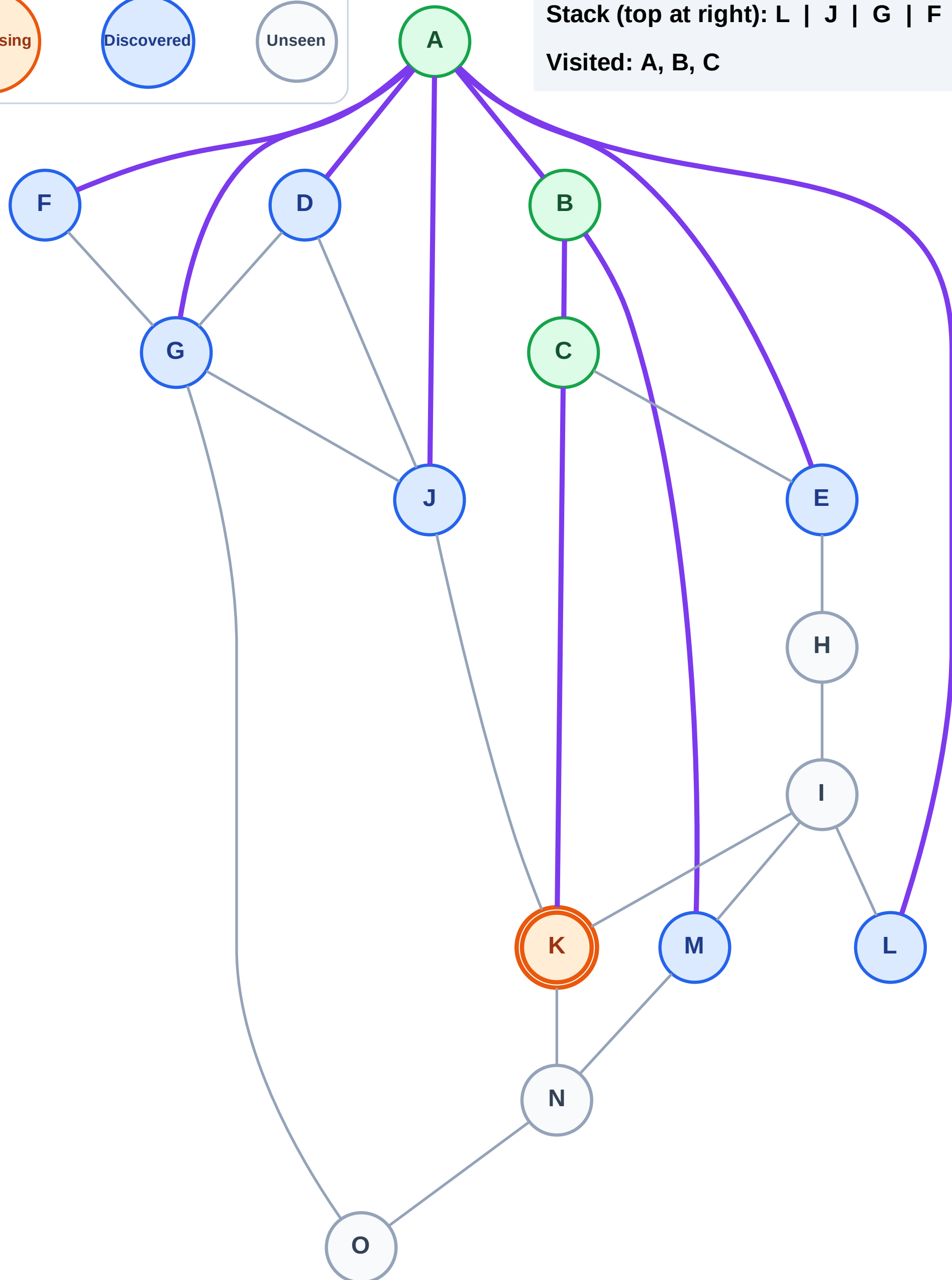
Processing

Discovered

Unseen

Stack (top at right): L | J | G | F | E | D | M

Visited: A, B, C



DFS Traversal

Step 9: TARGET NODE K REACHED! Path (3 hops): A → B → C → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): L | J | G | F | E | D | M

Visited: A, B, C, K

